



Review article

Development and therapeutic potential of adaptor-associated kinase 1 inhibitors in human multifaceted diseases

Xin Xin^{a,1}, Yue Wang^{b,1}, Lele Zhang^{a,1}, Dan Zhang^a, Leiha Sha^a, Ziyu Zhu^a, Xiaoyi Huang^a, Wuyu Mao^{a,*}, Jifa Zhang^{a,**}

^a Joint Research Institution of Altitude Health, State Key Laboratory of Biotherapy and Cancer Center, Department of Respiratory and Critical Care Medicine, Institute of Respiratory Health, National Clinical Research Center for Geriatrics, West China Hospital, Sichuan University, Chengdu, 610041, Sichuan, China

^b Leling Traditional Chinese Medicine Hospital, Leling, 253600, Shandong, China

ARTICLE INFO

Keywords:

AAK1
Small molecule inhibitors
Structure-activity relationships (SARs)
Anti-neuropathic pain
Broad-spectrum antiviral target

ABSTRACT

Adaptor-Associated Kinase 1 (AAK1), a Ser/Thr protein kinase, responsible for regulating clathrin-mediated endocytosis, is ubiquitous in the central nervous system (CNS). AAK1 plays an important role in neuropathic pain and a variety of other human diseases, including viral invasion, Alzheimer's disease, Parkinson's syndrome, etc. Therefore, targeting AAK1 is a promising therapeutic strategy. However, although small molecule AAK1 inhibitors have been vigorously developed, only BMS-986176/LX-9211 has entered clinical trials. Simultaneously, new small molecule inhibitors, including BMS-911172 and LP-935509, exhibited excellent druggability. This review elaborates on the structure, biological function, and disease relevance of AAK1. We emphatically analyze the structure-activity relationships (SARs) of small molecule AAK1 inhibitors based on different binding modalities and discuss prospective strategies to provide insights into novel AAK1 therapeutic agents for clinical practice.

1. Introduction

Adaptor-Associated Kinase 1 (AAK1), also known as AP2-associated protein kinase 1, belongs to the Numb associated kinase (NAK) family [1,2] and is a highly conserved Ser/Thr protein kinase [3,4]. AAK1 was originally named for its ability to phosphorylate the $\mu 2$ subunit of adaptor-related protein complex 2 (AP2) at Thr156 [5] and participate in clathrin-mediated endocytosis and secretion [6,7]. The AAK1 gene is located on the human chromosome 2p14 and encodes a 683 amino acid sequence [2]. As a protein kinase ubiquitously expressed in the central nervous system (CNS), chemical genetic analysis and mass spectrometry analysis confirmed that AAK1 is the substrate of nuclear Dbf2-related (NDR) kinase 1/2 [8] and PCTAIRE-1 (also known as cyclin-dependent protein kinase (CDK) 16) [9], which regulate dendritic branching and neurotransmitter transmission, respectively. Attenuating the expression of AAK1 can significantly control neuropathic pain and neurological disorders. Recent studies have shown that miRNA-384-3p can alleviate sevoflurane-induced neuronal apoptosis and neurological

damage by inhibiting AAK1 [10]. Therefore, AAK1 is considered as a potential anti-neuropathic pain target. In addition, AAK1 has a significant role in the control of numerous signal pathways and physiological processes [11–13]. AAK1 has been implicated in a variety of human disorders [14,15] including Alzheimer's disease (AD) [16], Amyotrophic lateral sclerosis (ALS) [17], Parkinson's disease (PD) [18], RNA virus entry and assembly [19], and others. Various vital functions of AAK1 throughout life have stimulated the production of AAK1 inhibitors (AAK1i) [20].

Notably, AAK1i has demonstrated remarkable clinical efficacy while treating COVID-19 [21–23]. Although there is currently no FDA-approved medication targeting AAK1, repurposed drugs have facilitated AAK1i research and development [24–27]. Sunitinib, recognized as AAK1i, is a pan-receptor tyrosine kinase (RTK) inhibitor licensed by the FDA for the treatment of cancer [28]. To date, many AAK1i have been developed; for example, LP-935509 alleviates persistent neuralgia in the mouse spinal nerve ligation (SNL) model [29]; inhibitors based on pyrrolo[2,3-*b*]pyridines skeleton increase resistance

* Corresponding author.

** Corresponding author.

E-mail addresses: wuyanmao1@sina.com (W. Mao), zjf298257@163.com (J. Zhang).

¹ These authors contributed equally to this work.

to DENV and EBOV viruses *in vitro* and human primary dendritic cells [30]. However, these drugs still have certain flaws, such as limited selectivity and unfixable dose dependence.

As the AAK1 target has become a hotspot in recent years, several inhibitors have entered the clinical phase. BMS-986176/LX-9211 is currently in development for diabetic peripheral neuropathic pain (ClinicalTrials.gov identifier: NCT04455633) and post-herpetic neuralgia (ClinicalTrials.gov identifier: NCT04662281) in Phase II clinical trials. Baricitinib is being integrated with Remdisivir in the Adaptive COVID-19 Treatment Trial 2 (ACTT-2) in Phase III clinical trial (ClinicalTrials.gov identifier: NCT04401579). Additionally, certain small compounds have recently demonstrated promise for medicinal use in preclinical studies. In a chronic constriction injury (CCI) model in rats, BMS-911172 significantly reduced neuropathic pain without causing dose-limiting adverse effects [29]; meanwhile, PHZ-OH employs AAK1 targeted inhibition to significantly lower caspase-11 activation and sepsis mortality [31].

This review emphasizes the structure, biological function, and disease relevance of AAK1. We summarize the structure-activity relationships (SARs) and binding modes of representative AAK1i. Meanwhile, we focus on the latest progress in AAK1i development and perspectives on relevant drug design strategies, contributing to future research into targeting inhibitors.

2. Structure and physiological function of AAK1

AAK1 comprises three domains: a highly conserved N-terminal Ser/Thr kinase domain (about 267 amino acids), an intermediate QPA-rich region (residues 312–630), and a C-terminal α -adaptin-interacting domain (AID) [2]. AAK1 bears a strong resemblance to other members of the NAK family in the kinase domain but differs in other domains [Fig. 1A]. The most relevant are AAK1 and BMP-2 inducible kinase (BIKE), which share 74% of their kinase domains and 50% of their total sequence [3]. Only three residues separate AAK1 from BIKE at the ATP binding pocket: two minor changes at the hinge region (Asp127 in AAK1 vs. Glu131 in BIKE, and Phe127 vs. Tyr132) and one significant change at the P loop (Ala58 vs. Ser63), which serves as the defining features to distinguish AAK1 from BIKE [Fig. 1B].

According to SDS-PAGE, it can be concluded that AAK1 mainly exists as a monomer rather than a dimer, with a typical bilobed domain [32]. The big-C helix (ASCH), a characteristic of the NAK family, is inserted by AAK1 into the C-terminal end of the activation region, directly affecting substrate binding and catalytic efficiency through conformational changes [2]. The N-terminal of the ASCH forms hydrogen bonds with Gln203 in the α E/ α F loop, while Leu207 and Ile243 interact in a similar manner. The glutamate (Glu229) side chain of the APE motif forms a salt bridge for activating the C-terminal of the loop. Furthermore, the arginine and tyrosine residues preceding the APE motif (Tyr225 and

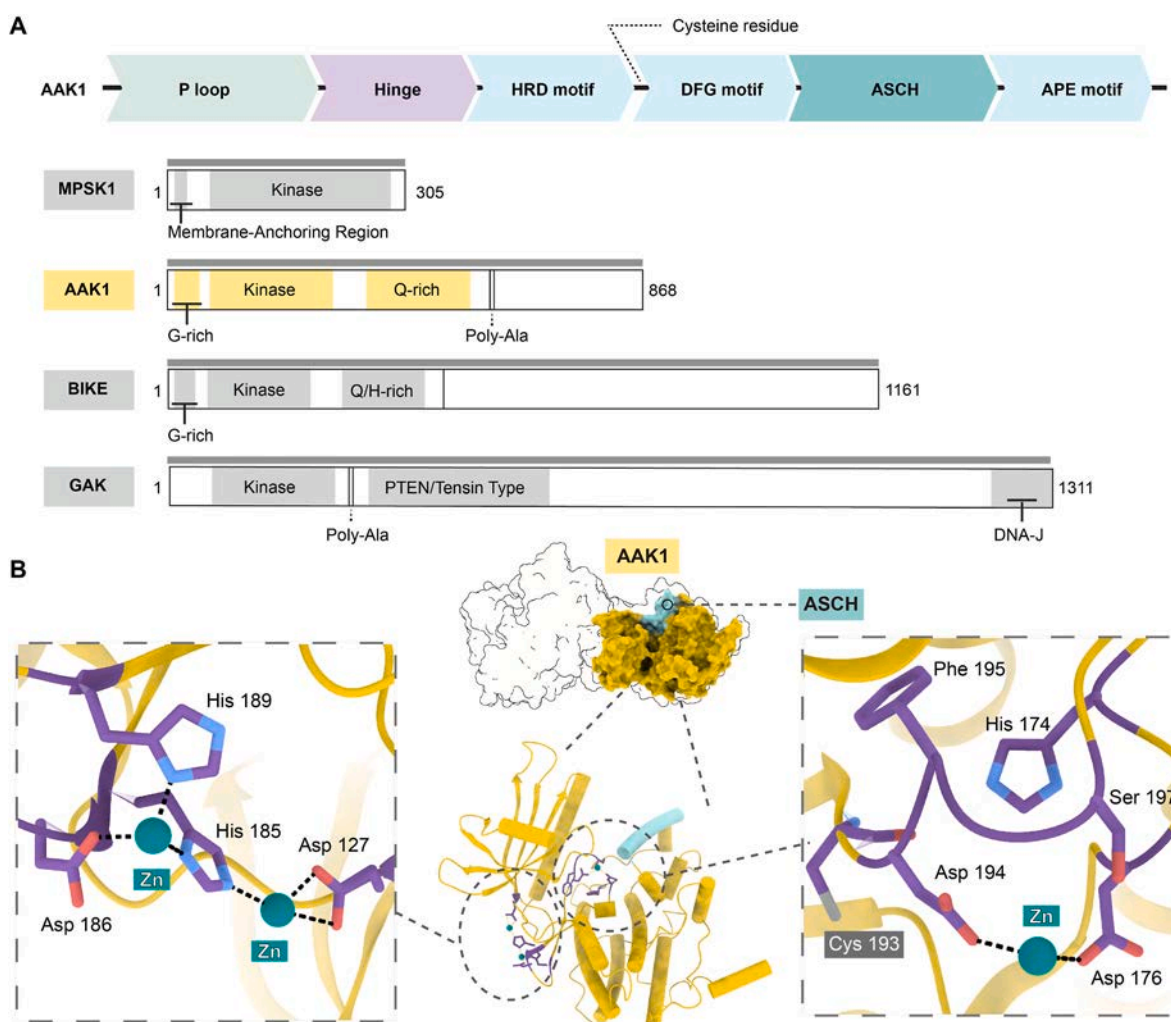


Fig. 1. Structure and dynamic of human AAK1. (A) Schematic diagram of the AAK1 kinase domain and domain organization comparison of human NAK family. (B) Details of AAK1 crystal structure (PDB ID: 4WSQ). ASCH (cyan): the activation C-terminal helix. The cysteine residue labeled in the grey rectangle provides a covalent binding site.

Arg226) appear to help immobilize the activation segment. Met94 from α -C helix, Tyr106 from β 4, and His174 and Phe195 from the HRD and DFG motifs constitute the R spine, suggesting that AAK1 is a constitutively active kinase. AAK1 features a unique phosphate-binding loop at the ATP site, called the “P loop,” which in AAK1 is Ala53-Ala59. The double-glycine motif in the loop allows it to adapt to various inhibitor scaffolds by changing the loop’s conformation. The cysteine (Cys193) in AAK1 provides an active site required for targeting covalent inhibitors [33], being a potential priority scheme to increase selectivity.

AAK1 is present at a high level in the cell membrane and cytoplasm. AAK1 accumulates in the presynaptic terminals of neuronal cells, while in non-neuronal cells it coexists with the clathrin-AP2 complex in clathrin-coated pits. AAK1 phosphorylates the μ 2 subunit of AP2 at Thr156 [34], which facilitates recognition and interaction with cell surface receptors to enhance endocytosis [35]. AAK1 is a NAK family homolog whose structure is most similar to BIKE [2] and whose function most closely resembles cyclin G-associated kinase (GAK) [36]. BIKE plays an important role in osteoblast differentiation [37], and GAK mediates the connection of clathrin to the plasma membrane and the trans-Golgi network (TGN), while AAK1 interacts with the α -subunit of AP2 in addition to promoting recognition on the membrane [5]. AAK1 is also involved in early/sorting endosomal receptor recycling of transferrin, however, overexpression of the long form of AAK1, encoding an additional clathrin-binding domain (AAK1L CBD2) impairs transferrin endocytosis [7]. Furthermore, overexpression of AAK1 inhibits low-density lipoprotein receptor-related protein (LRP), thereby preventing endocytosis, while reducing the number of phosphorylated subunits [4]. The high homology with mammalian Ark1/Prk1 suggests that AAK1 may play a role in coordinating the dynamics of cytoskeleton actin [38]. AAK1 has also been shown to be a crucial cytokine for viral infections and to be heavily engaged in neurodegenerative diseases, in addition to playing a significant role in the treatment of neuropathic pain [Fig. 2].

In PC12 cells, neuregulin 1 and receptor tyrosine kinase 4 (Nrg1/Erbb4)-dependent neurotrophic factor signaling was shown to be inhibited by AAK1. Inhibition of AAK1 will result in the sustained activation of downstream signaling by Erbb4 [39], leading to enhanced

Nrg1-mediated neurogenesis [Fig. 3]. Notch pathway regulates the cellular life course, which is linked to several human diseases, such as cancers [40]. AAK1 mediates Notch targeting to Rab5-positive endosomes to promote early endosome synthesis, directly influencing their membrane-tethered active form, and serves as an important cofactor in the Notch signaling pathway [41]. In addition, it ameliorates coagulation-induced organ dysfunction and lethality by inhibiting AAK1-mediated lipopolysaccharide (LPS) endocytosis to reduce caspase-11 pathway activation in sepsis models [31]. AAK1 modulates brain-derived neurotrophic factor (BDNF) to enhance paxillin-drebrin interactions and may promote neuronal morphogenesis as well as actin remodeling [42]. WNT activates AAK1 to promote endocytosis [Fig. 3], thus downregulating WNT signaling through a negative feedback loop [43]. Also, AAK1i stabilizes β -catenin and activates β -catenin-dependent WNT signaling [44], accelerating the transcription of target genes. As a member of the NAK family, AAK1 promotes dendritic morphogenesis while regulating early Numb function [45].

3. The role of AAK1 in human multifaceted diseases

AAK1 plays important roles in human multifaceted diseases, such as neuropathic pain, neurodegenerative diseases, and viral diseases. Inhibition of AAK1 activity can provide a variety of therapeutic outcomes. Besides, AAK1 has a synergistic effect with immune cells, which can also indicate the extent of tumor growth. Therefore, AAK1 becomes a potential and attractive therapeutic target for multifaceted human diseases.

3.1. Neuropathic pain

AAK1 gene knockout mice have an attenuated response to persistent pain, and AAK1i prevented mice from producing allodynia [7]. γ -aminobutyric acid (GABA) channel plays an important role in CNS [35,46], and μ 2-dependent endocytosis is extremely major for GABA_A Cl⁻ channels [47]. AAK1i reduces the amount of μ 2-contained GABA_A channel to achieve the analgesic effect, making AAK1 an anti-neuropathic pain target [Fig. 4]. Opioids are the most widely used drugs against neuralgia

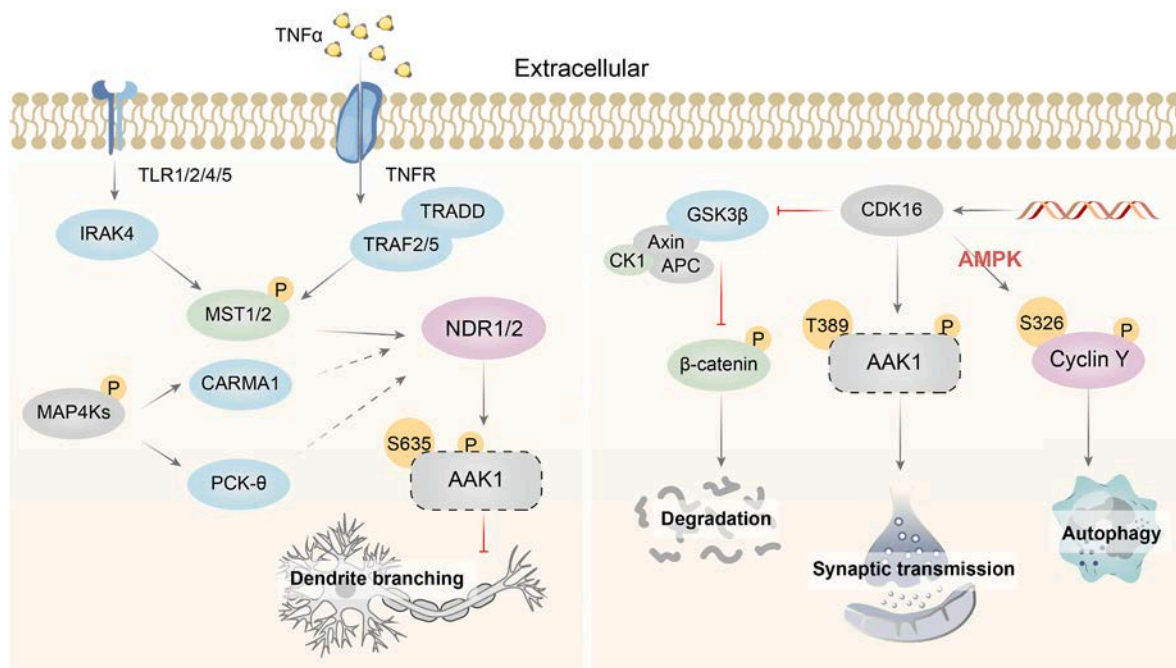


Fig. 2. The upstream genes of AAK1 and the signaling pathways involved in AAK1. Phosphorylated MST1/2 and MAP4Ks phosphorylates NDR1/2 and induce NDR1/2 to phosphorylate AAK1 at Ser635, which can inhibit the growth of dendrite branching. The abundance of miR-125b-5p can affect the activity of CDK16, which inhibits cell degradation and promote autophagy indirectly. CDK16 phosphorylates AAK1 at Thr389 to promote synaptic transmission.

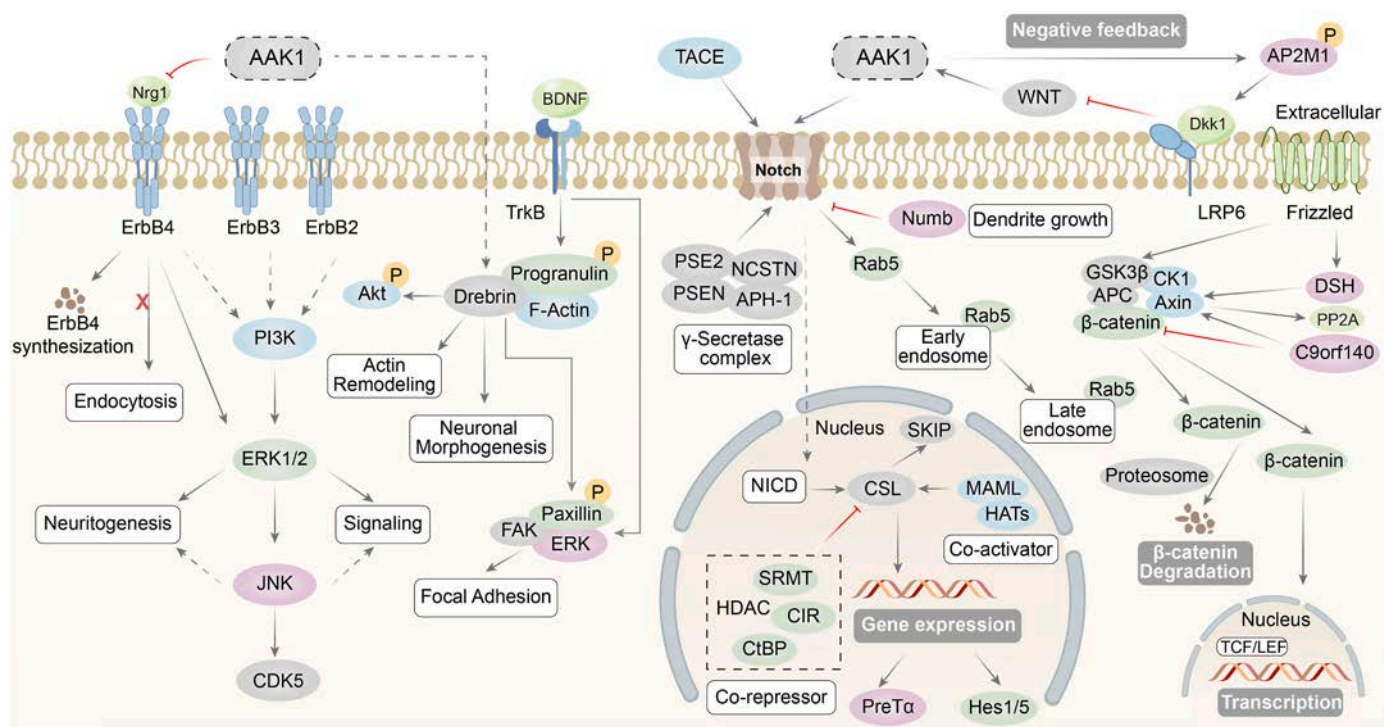


Fig. 3. The roles of AAK1 in multiple signal pathways. AAK1 inhibits Nrg1/ErbB4-dependent neurotrophic factor signaling, which activates ERK1/2 to enhance neurogenesis and signaling. Meanwhile, other ErbB receptors also have possibility of being inhibited by AAK1. AAK1 modulates BDNF to enhance paxillin-drebrin interactions and may promote neuronal morphogenesis and actin remodeling. AAK1 activates WNT pathway, which facilitates the Rab5 synthesis and transmission in the early and late endosome. Besides, the Numb protein activates AAK1 but inhibits Notch receptor, which indicates a dynamic balance. WNT activates AAK1 to phosphorylate AP2M1 (AP2-μ2) at Thr156, to promote clathrin-mediated WNT receptor (LRP6) endocytosis. However, the abundance of LRP6 also inhibits WNT, forming an AAK1-driven negative feedback loop.

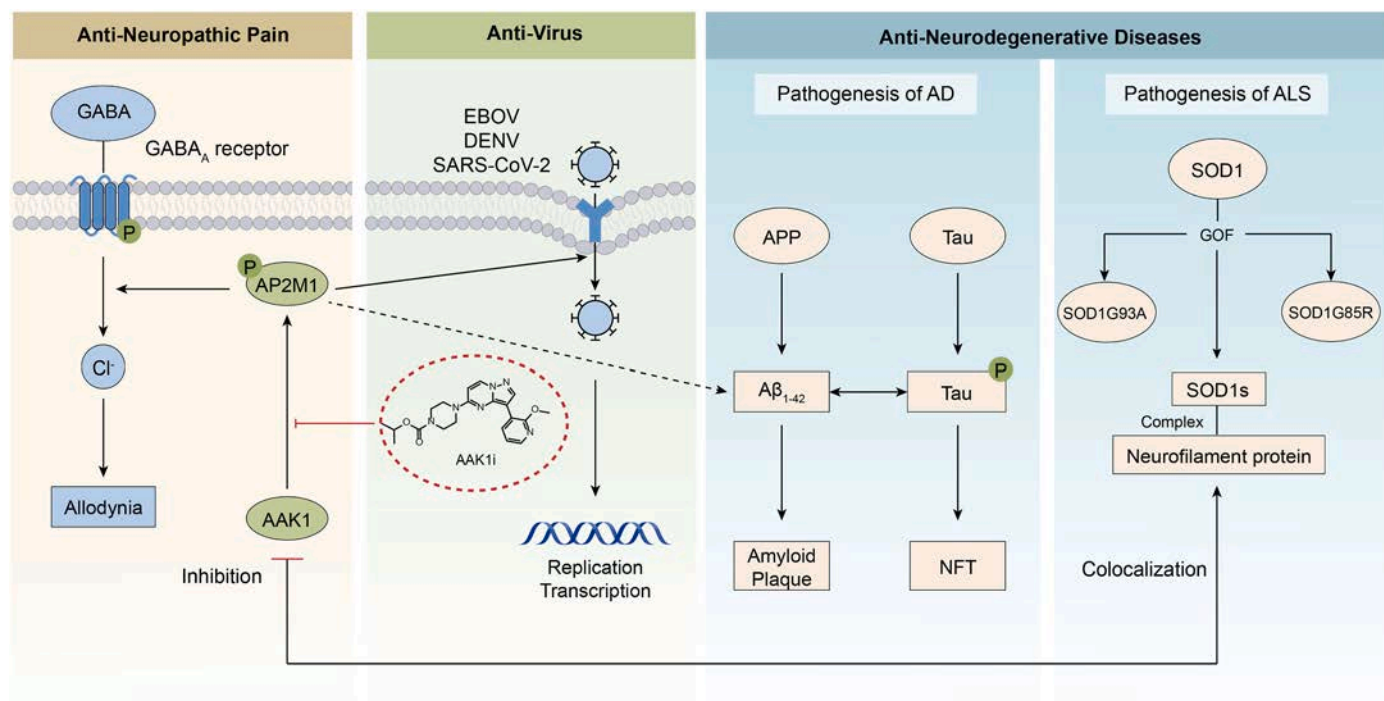


Fig. 4. AAK1 participates in the pathogenesis of multiple diseases. AAK1 can activate GABA_A Cl⁻ channels through phosphorylating AP2M1, which plays an important role in neuropathic pain sensation. AP2M1 is essential for virus that abducts the clathrin-mediated endocytic pathway to enter cells, and AAK1i decreases virus invasion by downgrading phosphorylated AP2M1 through inhibiting AAK1 activity. AAK1 can promote amyloid β (Aβ)-induced damage through AP2M1 in AD. AAK1 is colocalized with endosome and presynaptic protein markers, but it is wrongly located as an aggregate containing mutant SOD1s and neurofilament proteins in ALS, which complex can inhibit the activity of AAK1.

[48,49]. However, long-term administration is related to many adverse events, including immunosuppression, the disorder of self-reward, and neurohormone deficiency [50,51]. As an alternative, AAK1i avoids the inhibitory pathway of opioids, which it can also synergize with [29]. In the effective dose tests, mice treated with a reasonable dose of AAK1i did not show any obvious dyskinesia, implying its enormous research value [52].

3.2. Neurodegenerative diseases

AAK1 affects the alpha-synuclein aggregation and the resulting neurodegeneration in *Caenorhabditis elegans*, suggesting its great potential in combating neurodegenerative diseases [53]. **I.** AAK1 expression is enhanced in the cerebral cortex, hippocampus, and plasma of AD model mice [16]. Inhibition of endocytosis in AD models has prevented amyloid β ($A\beta$)-induced damage [54]. In $A\beta_{1-42}$ treated mice, AAK1 levels varied periodically, and in mice with cognitive impairment in the AD model, AAK1 expression was increased, indicating a negative correlation between AAK1 levels and cognitive function [Fig. 4]. **II.** There is a single nucleotide polymorphism (SNP) in intron 16 of the AAK1 gene on chromosome 2 [18], whose location and function are closely related to PD susceptibility genes [55,56]. AAK1 is associated with the age of onset of familial PD and affects the risk of PD by regulating lysosomal activity. Microarray studies of PD brains have shown differential expression of AAK1 [57], providing a new idea for disease surveillance. **III.** Approximately 20% of familial ALS (FALS) is caused by superoxide dismutase 1 (SOD1) gene mutation [58], and AAK1 can selectively interact with mutant SOD1s. Under normal physiological conditions, AAK1 partially colocalizes with endosome and presynaptic protein markers, but it is mislocalized as an aggregate containing mutant SOD1s and neurofilament proteins in ALS disease models. The level of AAK1 protein in ALS patients decreased, and the distribution of AAK1 was abnormal after onset [59]. AAK1 may be involved in ALS pathology in two ways: one is the loss of its regulatory function in the synaptic vesicle cycle, which may lead to presynaptic dysfunction as well as cell death; and the other is the presence of misfolding and aggregation in its expression pattern [17].

3.3. Viral diseases

Most viruses require internalization by endocytosis to penetrate and infect, and clathrin-mediated endocytosis is the most commonly used entry route for viruses [60]. After adhering to the cell surface and binding to specific receptors, the virus is loaded into clathrin-coated vesicles. Within the endosome, a decrease in pH can induce a conformational change in the viral envelope protein [61], allowing the virus to fuse with the endosomal membrane and release the virus into the host cell cytoplasm, thereby completing the invasion of the virus to the host cell. Therefore, AAK1 (cofactor of clathrin-mediated endocytosis) becomes a potential broad-spectrum antiviral target [Fig. 4].

AAK1 is extensively involved in the invasion mechanism of viruses: **I.** SARS-CoV-2 enters target cells through angiotensin-converting enzyme 2 (ACE-2)-mediated endocytosis [62], which is controlled by AAK1. AAK1 inhibition may interrupt virus propagation and intracellular assembly [63]. There was no significant difference in the expression level of AAK1 between patients with lung adenocarcinoma (LUAD) and normal subjects, however, the expression level of AAK1 was significantly decreased in COVID-19 patients infected with SARS-CoV-2 [64]. Baricitinib reduces inflammation by inhibiting Janus-kinase signaling and activator of transcription (JAK-STAT) pathways, and cooperatively inhibits AAK1 to reduce receptor-mediated viral endocytosis, thus achieving dual therapeutic effects [65,66]. **II.** Studies have shown that AAK1 regulates the assembly and entry steps in the HCV infection cycle [67,68]. AAK1 depletion leads to the loss of epidermal growth factor (EGF)-mediated enhancement of HCV entry and EGF receptor endocytosis [69]. **III.** RABV infection increased AAK1 transcription and

expression, whereas downregulation of AAK1 expression diminished the infection of RABV in N2a cells [70]. AAK1 knockout in HEK293 cells greatly inhibited cell infection with RABV, where AAK1 mRNA expression was reduced by 85% [71]. **IV.** DENV infection slightly increased the levels of AAK1 and GAK transcripts in a large number of Huh7 cells relative to uninfected cells, and AAK1 expression was related to the abundance of DENV in cells [72]. Although in the absence of AP2, endocytosis uptake of some receptors can be maintained by alternative clathrin adaptors, AP2 is necessary for optimal coat assembly and trafficking. AP2M1 (AP2- μ 2) is essential for any virus that abducts the clathrin-mediated endocytic pathway to enter cells, including influenza, adenovirus, and foot-and-mouth disease virus. Inhibiting AAK1 activity to down-regulate viral endocytosis has become an effective way to develop broad-spectrum antiviral drugs [73].

3.4. Other diseases

AAK1 expression and immune cell infiltration are closely related, and high AAK1 expression is associated with high $CD4^+$ T cell infiltration [64]. AAK1 inhibition can affect the expression of chemokine receptors and improve T cell trafficking *in vitro* and *in vivo*. Additionally, AAK1 has been identified in tumor models, providing a possible way for improving T cell aggregation in tumors. This target is also beneficial for applications using CAR-T cells and adoptive cell metastasis [74]. The abundance of AAK1 has emerged as one of the characteristic indices for prognostic prediction of diffuse large B-cell lymphoma [75]. Due to the depletion of AAK1 in RKO colon cancer cells, AAK1 emerged as a possible target for tumor therapy, suggesting that AAK1 may be involved in controlling apoptosis.

4. AAK1 inhibitors for the treatment of different human diseases

SARs-based drug design, radioactive displacement assay and a series of new screening techniques have facilitated the development of novel AAK1i [53,76]. We have categorized and summarized AAK1i for different disease types and chemical scaffolds, such as aryl amides, biaryl alkyl ethers, etc. At present, most of the known small molecule AAK1i belong to type I inhibitors, which target the conserved ATP-binding site in kinase activity and exhibit an Asp-Phe-Gly-in (DFG-in) conformation that facilitates phosphorylation transfer [77, 78]. They all provide structural references and ideas for the design and development of novel AAK1i and further lay the foundation for the development of AAK1 as a new therapeutic target.

4.1. AAK1 inhibitors for neuropathic pain

4.1.1. Aryl amides

Hydrogen interaction is an important factor in drug design. Moreover, chemical structures need to be selected that can bind to form hydrogen bonds, such as amides [79]. Aryl amides **1** (IC_{50} = 69 nM) was first identified by Richard et al. [80]. Subsequently, SARs-studies on the amide side chain was carried out [Fig. 5A]. Compound **2** has the best potency with isobutyl as the substituent, suggesting that the isobutyl side chain is the most suitable length side chain; the cyclopentyl-substituted analog **3** showed similar potency to **2**, indicating that the cyclopentyl group on the α -carbon is approximately equivalent to isobutyl, but the other cycloalkyl substitutions led to negative effects, implying that the space for the binding pocket to accommodate aliphatic side chains is limited. The introduction of the piperidine ring yielded compound **4**, whose metabolic stability (H/R/M) was substantially lower than that of **2** (89/75/64 vs. 60/42/2), and the end-group oxazole ring was superior to the imidazole ring because the nitrogen atom on the oxazole ring interacted with the hydrogen bond in the hinge area, with only compound **5** showing similar potency to **2**. Substitution of Me with CHF_2 on the benzene ring to obtain compound **6** (BMS-911172), showing higher activity in HEK293 cells (IC_{50} = 51 nM)

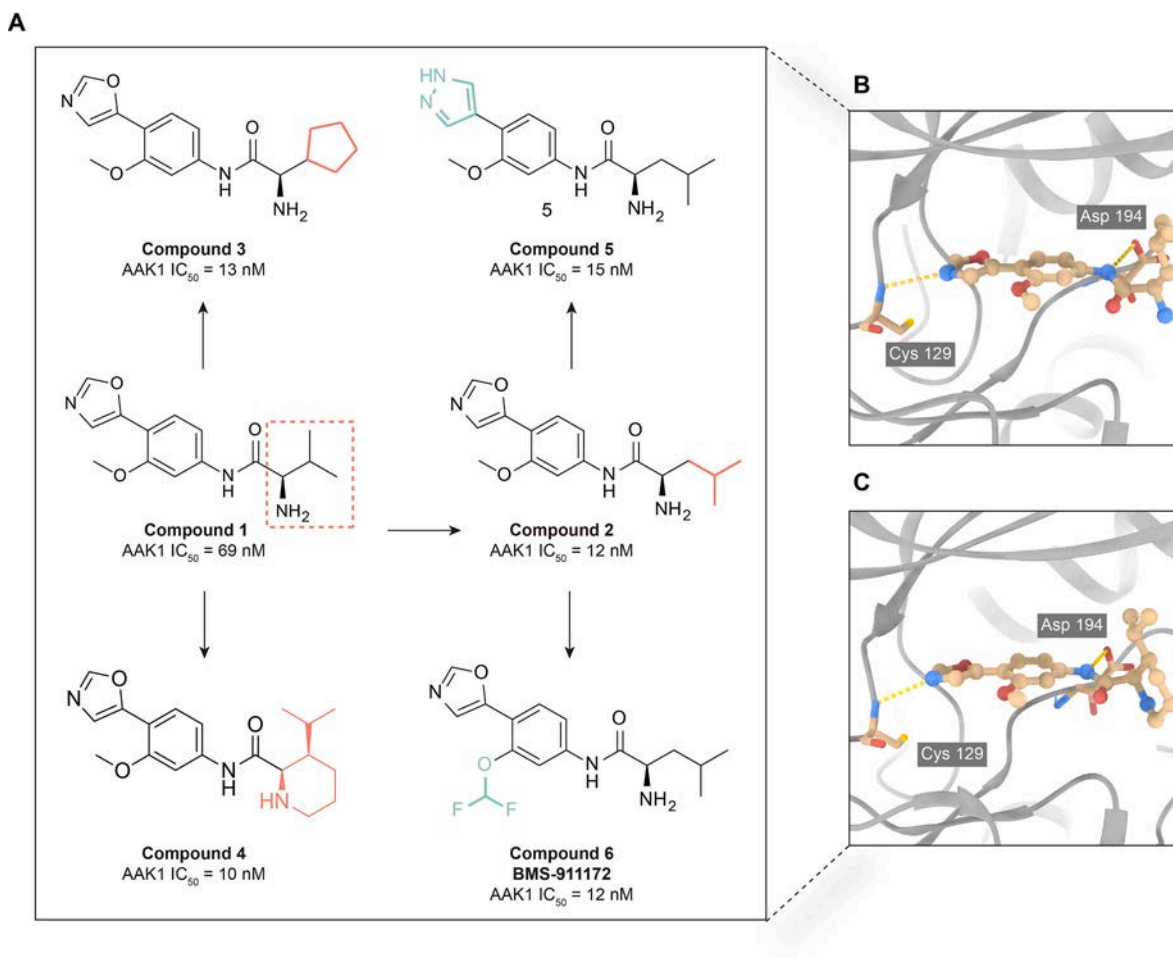


Fig. 5. (A) The structure-activity relationship of Aryl Amides. Binding mode of AAK1i to AAK1 structure: (B) compound 2 (PDB ID: 7LVH); (C) compound 4 (PDB ID: 7LVI).

and better potency in the plasma and brain of male C57BL6 mice. Compound 6 responded well only to NAK family members among 391 kinases tested, inhibiting μ 2 phosphorylation at concentrations consistent with those observed in AAK1 knockout mice. In addition, the rotarod model showed that it effectively avoided the dose-limiting side effects seen in the current therapeutic drugs and that the drug performed well in the mouse formalin-induced persistent pain model (Phase II) and the rat neuropathic pain CCI model.

The X-ray crystal structure (PDB ID: 7LVH) elucidates that compound 2 binds to the ATP site of AAK1 [Fig. 5B]. The nitrogen atom of the oxazole ring forms a hydrogen bond with the NH of the Cys129 backbone in the hinge region. The central phenyl group is sandwiched between Val60 of the P loop and Cys193 at the N-terminal of the DFG motif (Asp194-Phe195-Gly196); the equivalent interaction occurs between amide NH and side chain oxygen of Asp194 from DFG motif. The amine forms a hydrogen bond with the side chain oxygen of Asn181 and positions the isopropyl so that it is upward and close to the bottom of the P loop. Hydrophobic interaction between the isopropyl and small lipophilic pocket on P loop enhances drug potency and kinase selectivity. The crystal structure is consistent with the SARs findings, that is, the D-leucine-derived amide side chain exhibits stable hydrogen bonding, which promotes the bridge stability between the N-terminal region (P loop) and the C-terminal region (DFG motif) via the amine, amide nitrogen, and alkyl side chain with appropriate length. There is a critical hydrophobic interaction of methoxy with the upper Leu52 and Ala53 and the lower Leu183 from P loop, and the size of methoxy fits perfectly into this binding pocket, thus enhancing the effectiveness. By comparing the crystal structure [Fig. 5C], the decrease in the effectiveness of

compound 4 (PDB ID: 7LVI) is caused by the spatial conflict between the isopropyl and the protein. The ability of compound 6 (PDB ID: 7RJ8) to form hydrogen bonds with ATP binding site in the hinge region makes it a potential ATP-competitive small molecule kinase inhibitor, with other interaction relationships similar to compound 2.

4.1.2. Biaryl alkyl ethers

The generation of novel bis(hetero)aromatic ethers with high CNS permeability has given rise to fresh thought. After preliminary screening, Luo et al. obtained lead compound 7 [81], which was expected to develop an oral drug by switching amides for ethers to get rid of amide hydrogen bonds [Fig. 6B]. Improved cellular potency and CNS permeability were achieved [Fig. 6A]. Compound 8, obtained by replacing the amide with ether, resulted in a decrease in IC_{50} value and inhibition in most CYP isoforms due to the breakage of hydrogen bonds. The introduction of CF_3 (9) into the central benzene ring did not completely solve the problem in compound 8, and CYP3A4 inhibition remained. Insertion of a methyl group into the side chain amino group helped to reduce CYP3A4 inhibition, and CHF_2 substituted compound 10 performed well in terms of IC_{50} value as well as cellular potency. However, the single-point assay panel demonstrated its lowered kinase selectivity for AAK1 [82]. Substituting the central benzene ring with pyridine ring substantially improved selectivity, and retained other excellent properties. Compound 11 (BMS-986176/LX-9211) inhibited only 4 of the 396 kinases, while it was approximately 500-fold more selective for AAK1 than the others. The compound BMS-986176/LX-9211(11) at an oral dose of 0.3–1 mg/kg was effective in reducing the thermal hyperalgesia of CCI rats and showed excellent

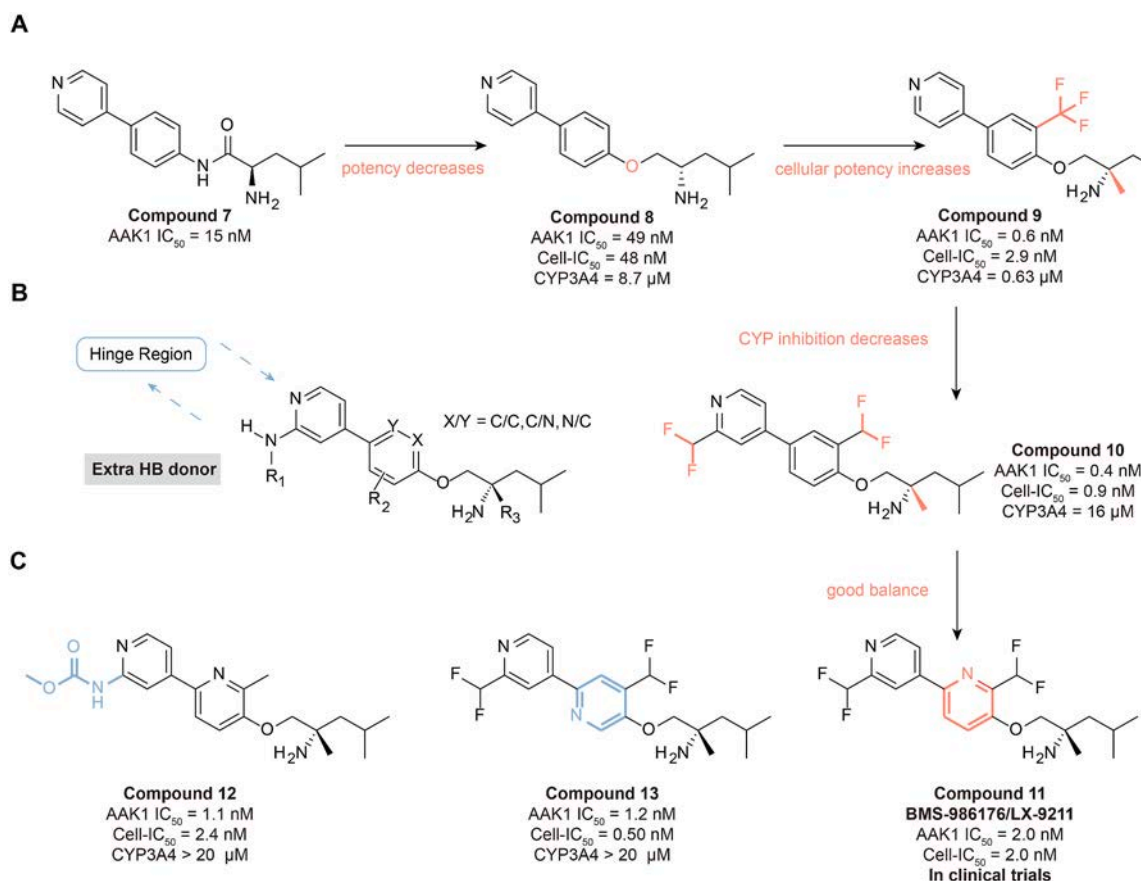


Fig. 6. (A) Design strategy of compounds 7–11. (B) Proposed AAK1i with an extra hydrogen-bond donor for better potency. (C) Modification on compound 11 for selective AAK1i. The Cell-IC₅₀ was tested in HEK293F cells.

spinal cord occupancy [83]. Its permeability to the CNS expanded further applications [84]. In addition, the preclinical results also showed that BMS-986176/LX9211(11) has a low potential for neuromotor injury and has entered clinical trials (NCT04662281 and NCT0445633).

Because of the excellent pharmaceutical properties of the biaryl alkyl ethers chemotype [Fig. 6B], the biaryl scaffolds were extensively explored and further optimized by Luo's group [85]. Compounds 12 and 13 [Fig. 6C] were obtained from BMS-986176/LX-9211 by substituting CHF₂ with an amide group as a critical modification and adjusting the conformation of the central pyridine ring, respectively. Compound 13, the central pyridine isomer of BMS-986176/LX-9211 (11), was 4-fold more potent than 11 *in vitro* and achieved benefit in the CCI rat model with lower plasma exposure. Compounds 12 and 13, at oral doses of 1–10 mg/kg, both exhibited good efficacy in the rat neuropathic pain model. And both possess high CNS penetration and excellent spinal cord targeting binding. These substances also show favorable physicochemical and oral pharmacokinetic (PK) properties. Nevertheless, the toxicity profile still must be considered, as both show toxicity to hepatocellular and other cardiovascular systems at high dosages.

4.1.3. Quinazoline derivatives

The strategy of optimizing bicyclic systems as modifiers to improve the effectiveness of inhibitors has been widely used [86,87]. Richard et al. synthesized a series of quinazoline analogs based on compound 6C4 by ring-forming reactions that would provide a new carrier [Fig. 7A], which can project substituents onto hydrophobic back-pocket substituents to improve the cellular efficacy of inhibitor and explore better AAK1i [88].

Substitution of the aryl amide of compound 14 with quinazoline gave compound 15, which showed a slight reduction in potency compared

with the preliminary product. Compound 16 obtained by inserting Et into the 4-position was able to enhance the potency. After enantiomeric separation of 11 by chiral supercritical fluid chromatography (SFC), it was found that the *S*-enantiomer [(*S*)-16] is more potent than the *R*-enantiomer [(*R*)-16]. Cyano substitution analog (17) was able to further improve the potency and became the optimal structure among 4-position alkyl ether and *N*-alkylamine analogs; these results indicate that the substituent at 4-position of quinazoline is able to bind to the small hydrophobic pocket in the AAK1 kinase domain and improve the potency. Based on the structure of 17, the IC₅₀ value of compound 18 could be further improved by inserting OMe at 7-position; the substitution of the terminal oxazole ring with a pyridine ring (19) resulted in an attenuation of potency. On the basis of compounds 18 and 19, quinazoline ring is replaced by quinoline ring to give compounds 20 and 21, both of which have lower IC₅₀ values and better effectiveness of *S*-configuration. In contrast, compound 20 has better cellular activity, kinase selectivity and metabolic stability. Compound (*S*)-20 down-regulated the phosphorylation of μ2 by 64% at a dose of 30 mg/kg (sc) with better brain penetration (a B/P ratio of 1.5), which deserves further optimization.

The crystal structure of *S*-16 with mouse AAK1 (PDB ID: 7RJ6) revealed a single hydrogen bond between the hinge-binding oxazole nitrogen and the backbone N of Cys129; the amine was shown to form hydrogen bonds with the side chain carboxylate of Asp194 from DFG motif and the side chain oxygen of Asn181 [Fig. 7B]. The hydrogen bond interaction between the side chain amine and the protein would maintain the orientation of the isopropyl to optimize the hydrophobic interaction between its distal isopropyl group and the hydrophobic pocket at the underside of the P loop region, as evidenced by the surface expression of the protein in the X-ray crystal structure. The quinazoline

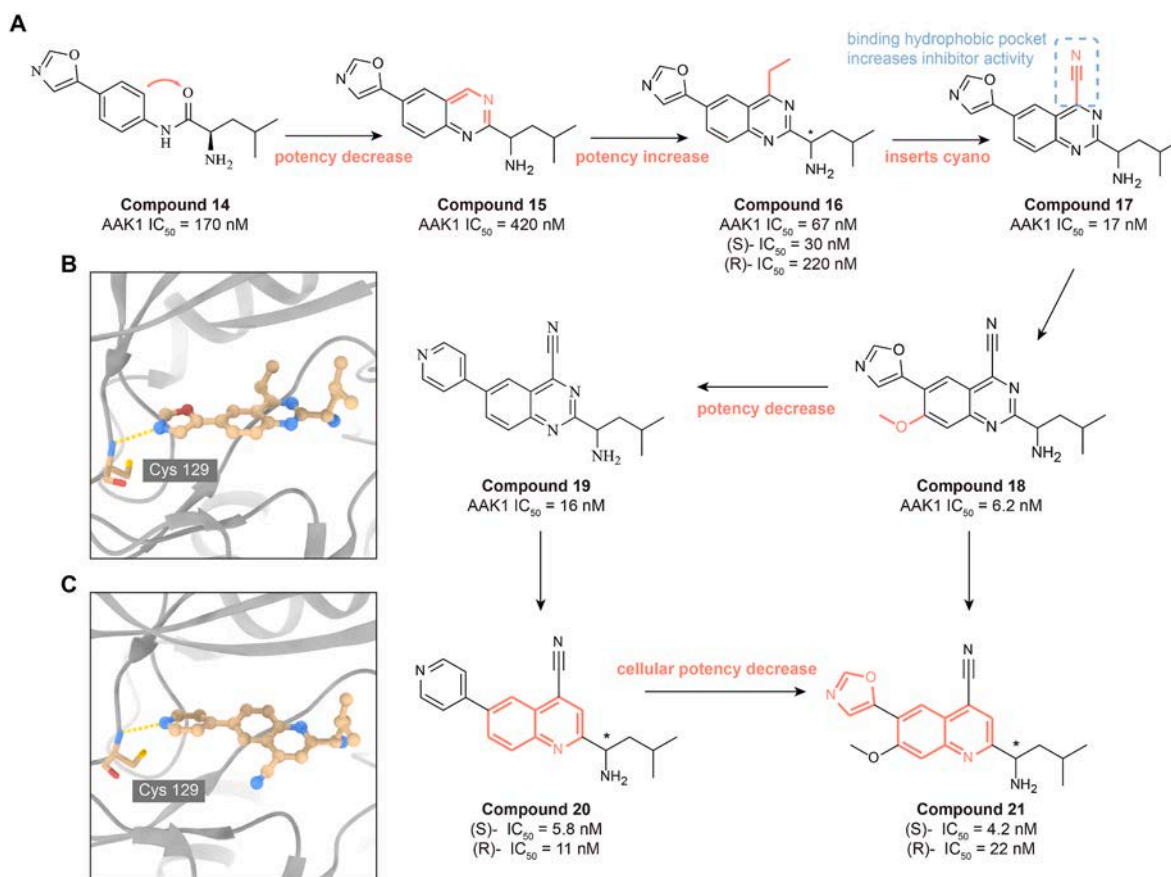


Fig. 7. (A) The structure-activity relationship of Quinazoline Derivatives. (B) X-ray crystal structure of AAK1 in complex with compound (S)-16 (PDB ID: 7RJ6), (C) compound (S)-20 (PDB ID: 7RJ7).

ring is located between the Val60 of P loop and Cys193 located at the N-terminal side of the DFG motif (Asp194-Phe195-Gly196), while the 4-ethyl substituent occupies a small hydrophobic pocket close to the gatekeeper residue (Met126). Site Map characterization revealed the hydrophobic-friendly region at the position of the C4 substituent, which may explain the increased potency of the ethyl analogy in (S)-16. The existence of a polar region adjacent to the hydrophobic region around the DFG motif, where the 4-cyano substituent of 17 reaches and may form a hydrogen bond or water-mediated hydrogen bond with the protein, explains the favorable activity of this analog. The crystal

structure of S-20 with AAK1 (PDB ID: 7RJ7) shows that the S-enantiomer of 20 binds in the same manner as (S)-16 [Fig. 7C]. The identical hydrogen-bonding interaction as well as hydrophobic interaction of the isopropyl group is present. The difference is that it replaces a bound water molecule and forms a water-mediated hydrogen bond with the amide backbone NH of Phe195, and this hydrogen bonding interaction contributes to the enhancement in the potency of the 4-cyano substituent analogs.

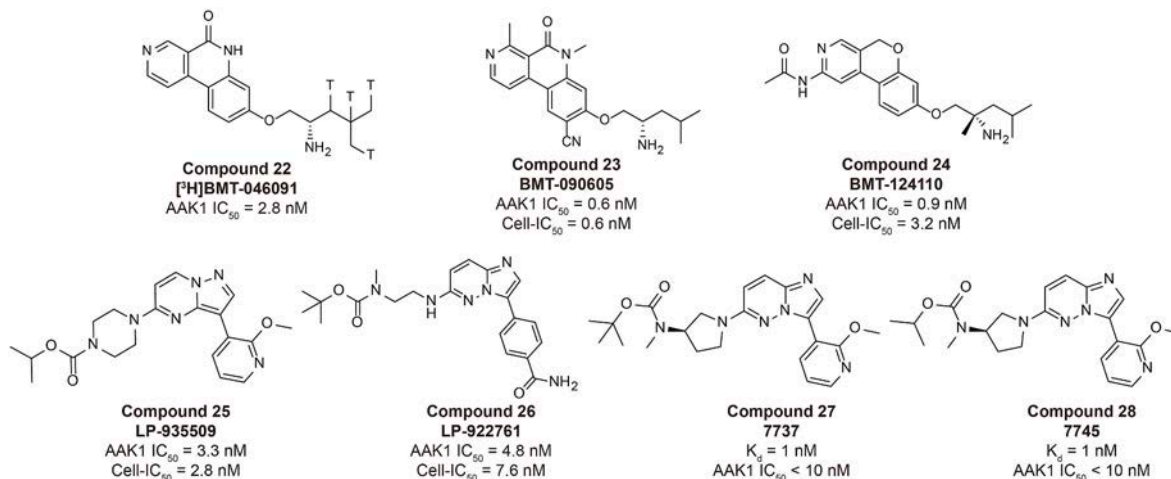


Fig. 8. Chemical structure of compounds 22–28. The Cell-IC₅₀ was tested in HEK293F cells.

4.1.4. Phenanthrene derivatives

[³H]BMT-046091 (**22**) is a potent and selective AAK1i [Fig. 8] and has been successfully used to measure the occupancy of brain targets following systemic AAK1 inhibition in rodents through radioligand [89]. [³H]BMT-046091 inhibited AAK1 phosphorylation of AP2M1 (IC₅₀ = 28 nM), which was absent in the brains of AAK1 knockout mice. In the spinal cord, [³H]BMT-046091 was enriched in the dorsal horn superficial layers, which was not altered by CCI surgery. Systemic administration of AAK1i occupied the AAK1 binding site, and occupation of this site correlated with anti-neuropathic pain effects, implying its potential in anti-neuropathic pain. In addition, BMT-090605 (**23**) and BMT-124110 (**24**) [Fig. 8] also exhibited intracellular AAK1 inhibition [29]. Experiments involving these two compounds indicated that AAK1 restraint in the spinal cord was essential for anti-neuropathic pain effects in rats. Oral administration of **24** did not show the same analgesic effect as intrathecal injection of **23** into rodents.

4.1.5. Pyrazolo[1,5-*a*]pyrimidine and Imidazo[1,2-*b*]pyridazine derivatives

In an enzyme assay using the structural domain of human AAK1 kinase, LP-935509 (**25**) inhibited phosphorylation of a peptide derived from the μ 2 protein (IC₅₀ = 3.3 nM), while inhibiting the binding to 13 kinases by more than 70% [50]. LP-935509 exhibited 100% oral bioavailability, a plasma half-life of 3.6 h, and high brain penetration. LP-935509 competed with ATP to reduce neuropathic pain behavior in mice and rats, but did not affect acute pain perception. In addition, it can be distributed throughout the body, matching the wide distribution of AAK1, moreover, the anti-neuropathic pain effect of LP-935509 was highly correlated with the occupancy value of the AAK1 binding sites. Radiographic autoradiography analysis showed a dose-dependent effect of LP-935509 on AAK1 binding sites in the dorsal horn of the spinal cord and no difference in potency between the spinal cord and forebrain regions. LP-922761 (**26**), an analogy of LP-935509, relieves neuropathic pain, but oral LP-922761 remains ineffective. In addition, compounds **27** and **28** with the same scaffold have been used to modulate AAK1 activity as potential therapeutic agents for neurological disorders [Fig. 8].

4.2. AAK1 inhibitors for viral diseases

To date, most antiviral therapies have been approved to target viral enzymes through a “one drug, one bug” approach. However, the low efficiency and high cost of drug development make it difficult to meet clinical needs on a large scale. In addition, virus-targeted medicines are typically less effective due to the development of drug resistance. This dilemma can be overcome by host-targeted therapies that focus on enhancing virus-dependent host resistance [90]. Because they provide the opportunity to repurpose drugs that have been shown to control particular host activities, host-targeted methods are appealing for the treatment of developing viral infections. Due to its crucial function in the viral life cycle, AAK1 is a potential target for the development of broad-spectrum antiviral drugs.

4.2.1. Pyrrolo[2,3-*b*]pyridine derivatives

DiscoverX analysis was used to evaluate the binding of 577 structural derivatives with a panel of 203 protein kinases. Researchers identified a pyrrolo[2,3-*b*]pyridine derivative (**29**) as a potent AAK1i (K_d = 53 nM) [91]. Compound **29** was selected as the lead compound, and a series of SARs studies [Fig. 9A] were performed [92]. Using 3,4-dimethoxyphenyl as the C5 substituent, the best potency was obtained, and the pyridine ring in compound **30** was also the best choice for the SARs studies of the N-Acyl moiety. On the basis of **28**, optimization with acetylene instead of amide gave compound **31**, but it has low selectivity and may be a better pharmacological tool to explore the role of AAK1 in viral infection. Compounds **29–31** effectively reduced the phosphorylation of AP2M1, and compounds **30** and **31** showed dose-dependent inhibition of DENV infection. Another nitrogen atom was inserted into the pyrrole part to produce pyrazolo[3,4-*b*]pyridine analog **32**, which was 100-fold less potent than **30**. When the pyridine moiety of compound **31** was replaced by a pyrazine ring, the pyrrolo[2,3-*b*]pyrazine analog **33** was obtained. It was endowed with a better AAK1 affinity than **31**, but showed greater cytotoxicity in Huh7 cells.

The crystal structure of AAK1 (PDB ID: 5L4Q) complexed with compound **29** shows that the pyrrolo[2,3-*b*]pyridine moiety is directly bound between two highly conserved residue side chains (Ala72 at the N-terminal β 2 and Leu183 at the C-terminal), in which the nitrogen atom combines with residues Asp127 and Cys129 in the hinge region,

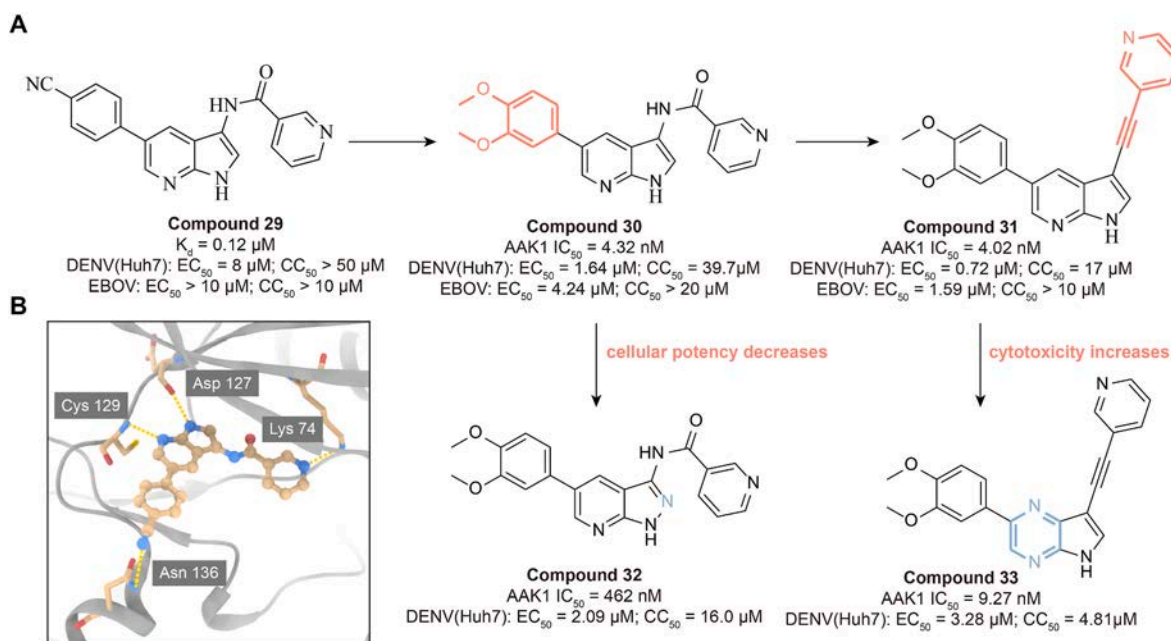


Fig. 9. The structure-activity relationship of Pyrrolo[2,3-*b*]pyridine Derivatives and X-ray crystal structure of AAK1 in complex with compound **29** (PDB ID: 5L4Q). The Cell-IC₅₀ was tested in Huh7 cells.

forming two hydrogen bonds [Fig. 9B]. The 4-cyanophenyl moiety is oriented toward the solvent, and the benzene ring is sandwiched between Gly132 of the hinge region and Leu52 of β 1. The nitrogen atom of the cyano group interacts with Asn136 on the side chain to form a polar interaction. The nicotinamide moiety binds to the gatekeeper residue Met126 and forms a hydrogen bond with Lys74 from the side chain; another highly conserved residue usually links the phosphate of ATP to the α -C helix. Compound **29** also interacts with two water molecules: one behind the ATP pocket and the other in front of the adenosine binding site below Val60 of the β 1 strand.

4.2.2. Repurposing existing drugs

Drug repurposing is a beneficial therapeutic strategy used commonly. At present, several FDA-approved drugs have good inhibitory effects on AAK1 [Fig. 10], which provides a solution for the development of new drugs and treating unknown viruses. Sunitinib (**34**) is an oral drug approved by the FDA for the treatment of renal cell carcinoma and imatinib-resistant gastrointestinal stromal tumor [93]. Liver tissue lysates collected from AG-B6 mice after dosing show that the combination of Sunitinib (**34**) and Erlotinib (**35**) caused dose-dependent AAK1 inhibition, confirming AAK1 as an antiviral target [28]. At the same time, the combination of the two drugs achieved sustained inhibition of systemic infection in DENV-infected IFN- α/β and IFN- γ receptor-deficient mice. In addition, Sunitinib deters RABV infection by blocking the kinase activity of AAK1. However, AAK1 overexpression completely reversed the antiviral effects of Sunitinib and Erlotinib [72], and the safety of this therapy still needs further exploration. Baricitinib (**36**) was originally an oral Janus-associated kinase (JAK) 1/2 inhibitor approved for the treatment of rheumatoid arthritis (RA) [94]. Studies have shown that it can inhibit AAK1 by regulating the internal transport-related cell surface protein ACE-2, and significantly reduce the SARS-CoV-2 viral load, inflammatory markers, and IL-6 levels in patients with bilateral COVID-19 pneumonia, further confirming its potential antiviral activity [23,95]. Fedratinib (**37**) is an oral small molecule ATP-competitive inhibitor of JAK2 and FMS-like tyrosine kinase 3 (FLT3; CD135; STK1; FLK2) with potential antitumor activity [96]. The high-affinity inhibition of AAK1 potentially makes it a potent precursor for the development of AAK1i. Cynaroside (**38**) is a β -D-glucoside flavonoid that interacts with AAK1 active site residues Leu52, Val60, Phe128, Cys129, Gln133, Leu183 and Asp194 with high affinity. Current studies have shown that Cynaroside is effective in the treatment

of patients with COVID-19 [97]. Midostaurin (**39**), a Flt3 inhibitor, exhibits anti-HCV activity by inhibiting AAK1 [26]. As an alternative therapy, drug repurposing can not only reduce the risk of failure and R&D costs, but also improve the success rate of treatment through drug synergy [98].

Likewise, kinase inhibitors are a powerful group of therapeutic agents with a tendency to block several targets [99,100]. Quantitative analysis of the selectivity of kinase inhibitors indicated there are good affinity activities between AAK1 and the current group of kinase inhibitors (compounds **40–51**) which merit further exploration [Fig. 11]. However, drug selectivity is another important factor to be considered, not just high affinity. For example, the c-Jun N-terminal kinase (JNK) inhibitor SP600125 (**52**) interacts strongly with BIKE and AAK1 but exhibits selectivity for 300 kinases [101]. Notably, the present and upcoming findings suggest that RTK inhibitor is possible AAK1i [102].

4.3. AAK1 inhibitors for other diseases

PHZ-OH (**53**) may act as a favorable small molecule for AAK1i, which may facilitate the development of highly potent agents for the treatment of sepsis-associated coagulation, multiple organ dysfunction, and death in the future [31]. Molecular docking results showed that AAK1 had a high binding energy to PHZ-OH, and surface plasmon resonance (SPR) results revealed a high affinity ($K_d = 4670$ nM) between PHZ-OH and AAK1. Both AAK1 knockout and PHZ-OH administration alone can significantly inhibit outer membrane vesicle (OMV)-induced pyroptosis in WT/Gbp2-deficient cells [103], thus the results suggest that PHZ-OH can prevent AAK1-mediated LPS internalization for caspase-11 activation [104].

4.4. Other AAK1 inhibitors

In addition to the above large number of AAK1i used to treat diseases, other different types of compounds also exhibit AAK1 inhibitory effects [Fig. 12].

4.4.1. Nature products derivatives

RNAi data suggest that AAK1 is a candidate target for the natural protein kinase C (PKC) inhibitor K252a (**54**) [39], which has a greater affinity for K252a compared to K252a-Me (**55**), suggesting that insertion of methyl groups is sterically unfavorable and requires higher energy to

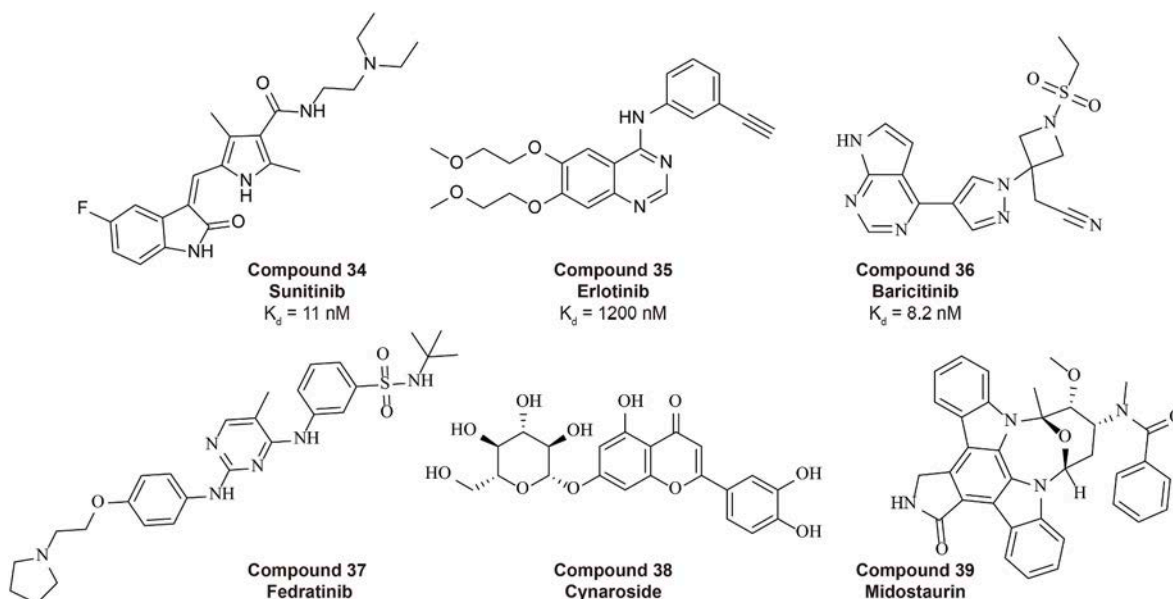


Fig. 10. Chemical structures of compounds **34–39**. FDA-approved drugs and clinical drugs which inhibit AAK1.

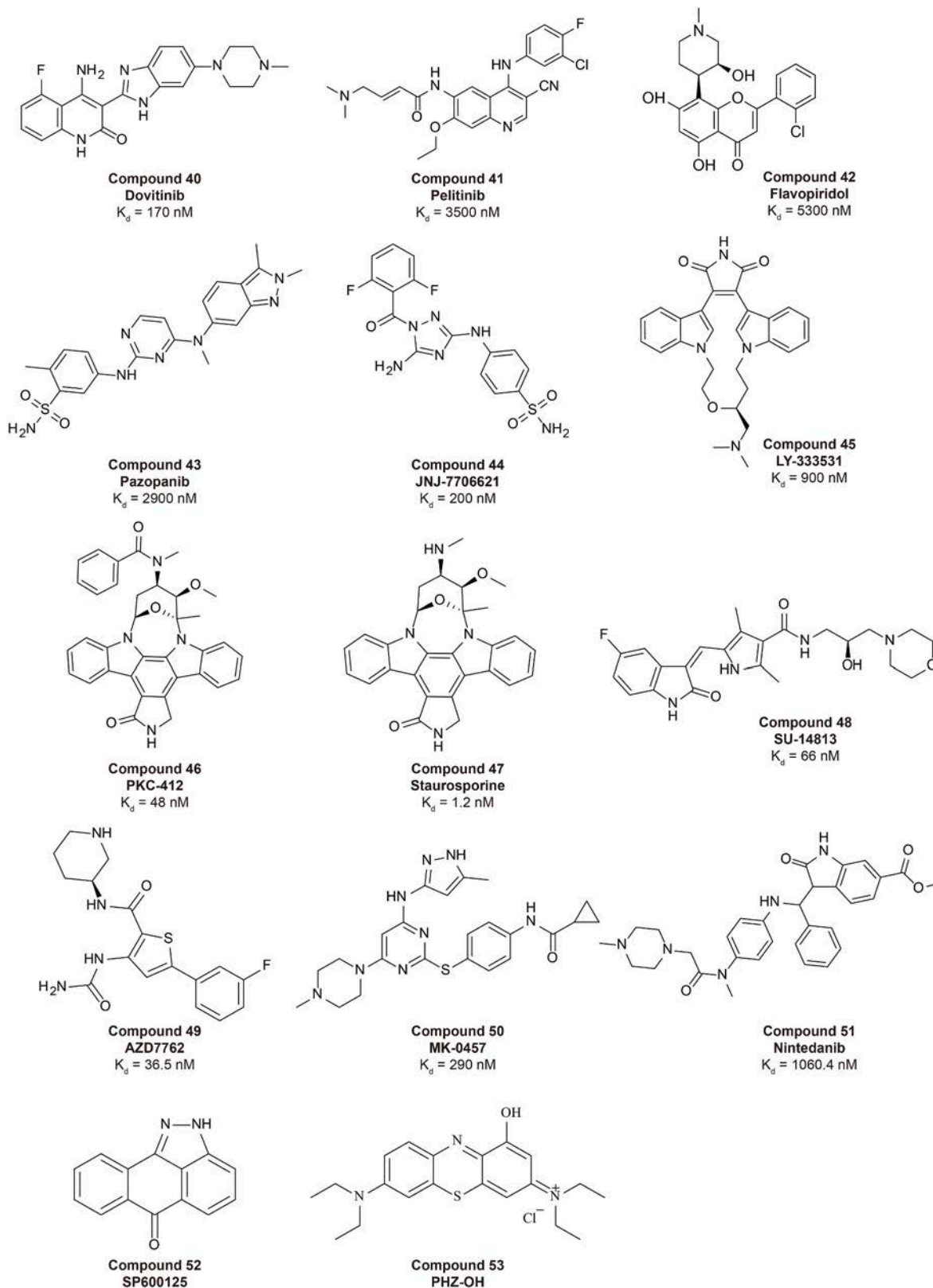


Fig. 11. Chemical structures of compounds 40–53. These compounds have a high affinity with AAK1.

adjust the conformation. (5Z)-7-oxozeaenol (56) is a natural anti-protozoal compound, which irreversibly binds to various kinases [105]. Incubation of (5Z)-7-oxozeaenol with recombinant AAK1 protein resulted in 132 Da changes in the observed protein molecular weight, confirming its covalent binding with AAK1 at Cys193.

4.4.2. Aminopyrimidine derivatives

Aminopyrimidine represents a novel chemical probe of AAK1/BIKE and is capable of inhibiting different off-target kinases [106]. Compounds 57 and 58 were found to be potent inhibitors of NAK family members (AAK1, BIKE and serine/threonine kinase (STK) 16) [107]. In

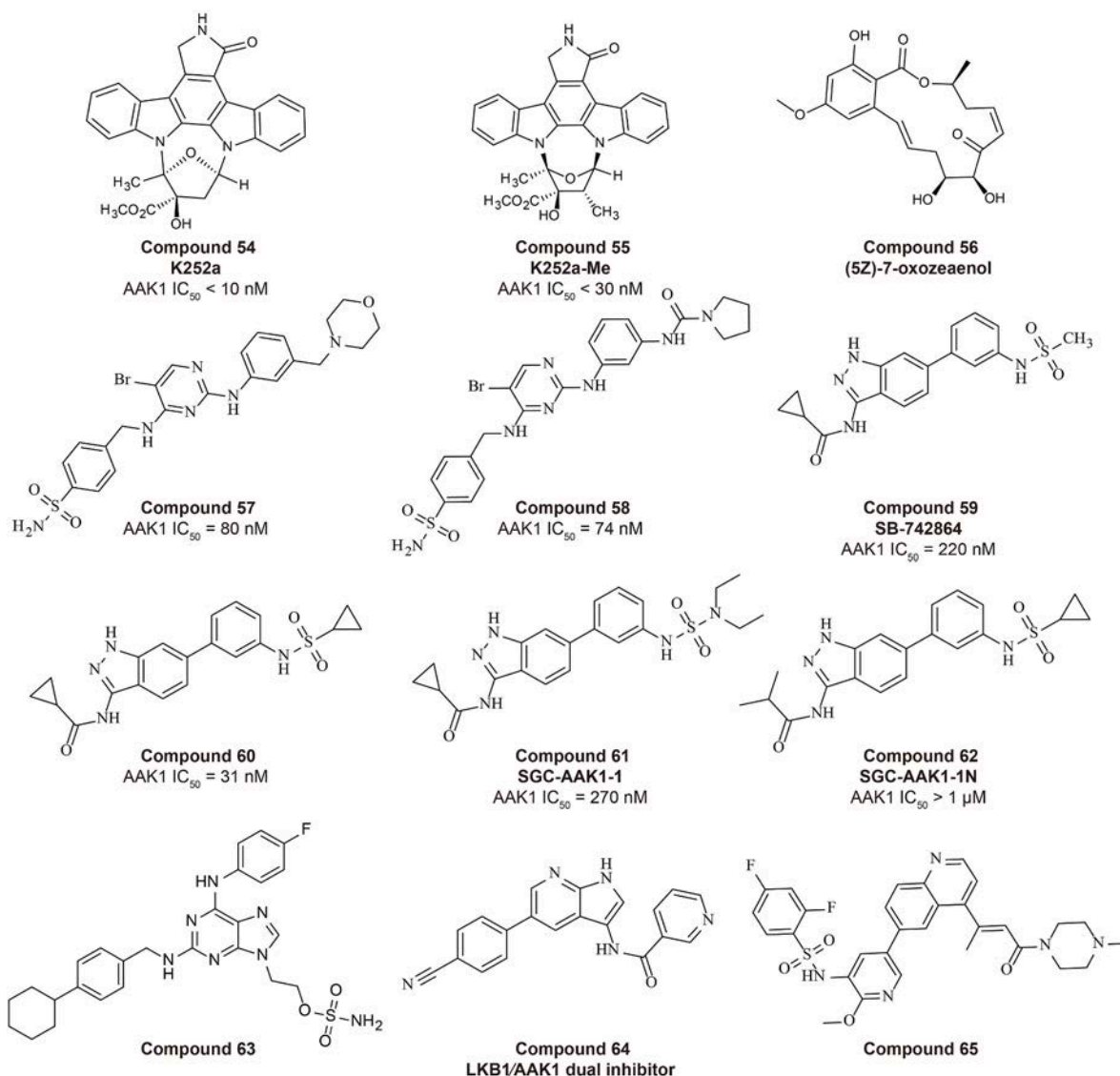


Fig. 12. Chemical structures of compounds 54–65. These compounds with potential to be optimized as AAK1i.

addition, the introduction of the bromine group enhanced the potency of the inhibitor. It was found that the halogen bond is very important for the ligand efficiency of AAK1 [108].

4.4.3. Indazole derivatives

The indazole SB-742864 (59) was identified as a semi-selective inhibitor of AAK1 [109], and its affinity for AAK1 was well confirmed in the split-luciferase assay. The indazole 3-substituent formed polar interactions with the kinase hinge region, with small hydrocarbon chains and rings or alkylamines on sulfonamides showed better AAK1 inhibition. Iterative optimization led to an analog 60 of SB-742864, which effectively and selectively inhibited AAK1 and BIKE. Further optimization to improve selectivity resulted in SGC-AAK1-1 (61), which showed good selectivity for a panel of 406 wild-type human protein kinases. Only a few other kinases were inhibited with dissociation constant values less than 30 times in the range of AAK1 [110]. Like the lead compound 60, SGC-AAK1-1 inhibited AAK1-dependent AP2M1 phosphorylation in HEK293T cells [43]. Opening the cyclopropyl carboxamide of compound 60 yielded the 2-methylpropionamide analog 62 (SGC-AAK1-1N), which had no significant AAK1 or BIKE inhibitory activity. Meanwhile, SGC-AAK1-1 (61) is an excellent probe candidate

targeting AAK1 and BIKE due to its excellent performance in various indices [110,111].

4.4.4. Pan-AAK1 inhibitors

Compound 63, a 2,6,9-heterotrisubstituted purine, was identified as a potential inhibitor of multiple myeloma (MM) tumor cells. It may target the JAK family, ABL proto-oncogene 1 (ABL1), and AAK1, and 52% of AAK1 treated with compound 63 was significantly inhibited according to quantitative PCR (qPCR) screening [112]. In addition, compound 64 is a dual-target inhibitor of liver kinase B1 (LKB1) and AAK1 that has been reported. Compound 65 can significantly inhibit the phosphoinositide 3-kinase (PI3K) family of kinases, and the KINO-MEScan™ assay of DiscoverRx showed that it has a strong hit against AAK1 [113], which may become a hit compound targeting AAK1.

5. Conclusions and perspectives

AAK1 controls clathrin-mediated endocytosis, which is prevalent throughout the body and particularly active in the CNS. As a cofactor for activated GABA_A, it is widely involved in neurotransmission. The primary target of antiepileptic drugs is GABA_A [114], which is the

predominant GABA receptor [115]. AAK1 may diminish the expression of GABA_A downregulated by abnormal autophagy in epilepsy to a certain extent, being the key to developing novel clinical drugs. Furthermore, AAK1 is the downstream target of NDR1/2 and CDK16, while NDR kinases [116,117] and CDK kinases [118,119] play a role in cell proliferation and cell death, making AAK1 an effective target for the treatment of various cancers and tumors. Meanwhile, AAK1 can label the accumulation of T cells and has potential synergistic effects [63]. AAK1 is involved in the regulation of the Notch pathway and the WNT pathway, which are tightly associated with cancer [120–122]. In particular, WNT-directed therapy represents an important developmental therapeutic area. These pharmacological reports have further contributed to the investigation of how AAK1 is involved in the life of cancer cells [123], and provide an alternative option for cancer treatment. At the same time, AAK1 has recently been discovered as a potential therapeutic target to cure COVID-19 after SARS-CoV-2 infection [124], and a series of clinical trials have been conducted. In the treatment of SARS-CoV-2-induced pneumonia, patients were given Baricitinib, in which clinical and respiratory parameters were significantly improved [125]. In addition to the above-mentioned causes, the causes of nerve pain include damage to the peripheral nervous system, such as tumors, which can also be caused by HIV infection [126,127]. Involved in tumors and HIV, AAK1 is also an important component of the CNS that regulates neurotransmitters and dendritic growth, indicating that AAK1 is worthy of further exploration in the treatment of neuropathic pain.

We summarize the progress of small molecule AAK1i in recent decades, most of which are type I small molecule inhibitors, in other words, they are ATP-competitive inhibitors that still suffers from poor selectivity [128,129]. Type III inhibitors (binding outside the ATP-binding site) can overcome the problem of weak selectivity, and their ability to stay on target for a long time provides space for improving efficacy [130]. There are some type III inhibitors designed and approved by FDA, which show great strengths [131], such as Cobimetinib (a mitogen-activated extracellular signal-regulated kinase (MEK) 1/2 inhibitor, FDA) [132] and ARQ902 (a protein kinase B (PKB) inhibitor) [133]. The specific ASCH sequence in AAK1 provides the possibility of developing type III inhibitors [134]. Meanwhile, Cys193 in AAK1 provides a binding site for the development of covalent inhibitors, which can well address kinase resistance [135]. Moreover, different chiral conformations of small molecules can bring great differences in potency [136]. In the case of AAK1, the amino side chain on the S-configuration carbon atom brings better potency. Cytotoxicity is also a non-negligible factor. Actively developing natural product-derived drugs is conducive to the expansion of new drug scaffolds and the development of human-tolerated small molecule inhibitors, providing a basis for their clinical application [137].

Although AAK1i is being extensively studied, there is no commercially available inhibitor. When it comes to the development of AAK1i, there are three major challenges: **I.** It is difficult to achieve high selectivity of kinase inhibitors due to the high similarity of AAK1 to other members of the NAK family; **II.** Although AAK1i can effectively bind to substrates through hydrogen bonds and hydrophobic interactions, they still suffer from poor metabolic stability and poor blood-brain barrier (BBB) permeability; **III.** We need to clarify the variable configuration sites of AAK1 to promote the development of specific inhibitors. In summary, further in-depth study on the interaction between AAK1 typical structural domain binding sites and inhibitors will provide a series of support and guidance for the development of more effective AAK1i [123]. The biological function of AAK1 as well as the high-resolution structure of the protein need to be further investigated.

In the past decades, a small number of inhibitors against AAK1 have been reported. Although their development may face many challenges, it is highly expected that more AAK1-specific inhibitors with favorable potency will be discovered, and allow the expansion of their clinical applications to human multifaceted diseases.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

This review was supported by National Natural Science Foundation of China (81903502, 22277083), and Natural Science Foundation of Sichuan Province (2022NSFSC1365, 2022NSFSC1187).

References

- [1] E.B. Sorensen, S.D. Conner, AAK1 regulates Numb function at an early step in clathrin-mediated endocytosis, *Traffic* 9 (2008) 1791–1800.
- [2] F.J. Sorrell, M. Szklarz, K.R.A. Azeez, J.M. Elkins, S. Knapp, Family-wide structural analysis of human numb-associated protein kinases, *Structure* 24 (2016) 401–411.
- [3] E. Smythe, K.R. Ayscough, The Ark1/Prk1 family of protein kinases, *EMBO Rep.* 4 (2003) 246–251.
- [4] T. Takahashi, T. Furuchi, A. Naganuma, Endocytic Ark/Prk kinases play a critical role in Adriamycin resistance in both yeast and mammalian cells, *Cancer Res.* 66 (2006) 11932–11937.
- [5] S.D. Conner, T. Schroter, S.L. Schmid, AAK1-mediated mu 2 phosphorylation is stimulated by assembled clathrin, *Traffic* 4 (2003) 885–890.
- [6] S.D. Conner, S.L. Schmid, Identification of an adaptor-associated kinase, AAK1, as a regulator of clathrin-mediated endocytosis, *JCB (J. Cell Biol.)* 156 (2002) 921–929.
- [7] D.M. Henderson, S.D. Conner, A novel AAK1 splice variant functions at multiple steps of the endocytic pathway, *Mol. Biol. Cell* 18 (2007) 2698–2706.
- [8] S.K. Ultanir, N.T. Hertz, G. Li, W.-P. Ge, A.L. Burlingame, S.J. Pleasure, K. M. Shokat, L.Y. Jan, Y.-N. Jan, Chemical genetic identification of NDR1/2 kinase substrates AAK1 and Rabin8 uncovers their roles in dendrite arborization and spine development, *Neuron* 73 (2012) 1127–1142.
- [9] S.N. Shehata, M. Deak, C. Colodet, S.F. Spiegl, S. Geley, D. Sumpton, K. Sakamoto, Identification of novel PCTAIRE-1/CDK16 substrates using a chemical genetic screen, *Cell. Signal.* 59 (2019) 53–61.
- [10] Y. Chen, X. Gao, H. Pei, miRNA-384-3p alleviates sevoflurane-induced nerve injury by inhibiting Aak1 kinase in neonatal rats, *Brain and Behavior*, 12 (2022) e2556.
- [11] G.J. Doherty, H.T. McMahon, Mechanisms of endocytosis, *Annu. Rev. Biochem.* 78 (2009) 857–902.
- [12] S. Watanabe, E. Boucrot, Fast and ultrafast endocytosis, *Curr. Opin. Cell Biol.* 47 (2017) 64–71.
- [13] H.T. McMahon, E. Boucrot, Molecular mechanism and physiological functions of clathrin-mediated endocytosis, *Nat. Rev. Mol. Cell Biol.* 12 (2011) 517–533.
- [14] P. Cossart, A. Helenius, Endocytosis of viruses and bacteria, *Cold Spring Harbor Perspect. Biol.* 6 (2014).
- [15] A.I. Krahn, C. Wells, D.H. Drewry, L.K. Beitel, T.M. Durcan, A.D. Axtman, Defining the neural kinome: strategies and opportunities for small molecule drug discovery to target neurodegenerative diseases, *ACS Chem. Neurosci.* 11 (2020) 1871–1886.
- [16] X. Fu, M. Ke, W. Yu, X. Wang, Q. Xiao, M. Gu, Y. Lu, Periodic variation of AAK1 in an A beta(1–42)-induced mouse model of Alzheimer's disease, *J. Mol. Neurosci.* 65 (2018) 179–189.
- [17] B. Shi, S.D. Conner, J. Liu, Dysfunction of endocytic kinase AAK1 in ALS, *Int. J. Mol. Sci.* 15 (2014) 22918–22932.
- [18] J.C. Latourelle, N. Pankratz, A. Dumitriu, J.B. Wilk, S. Goldwurm, G. Pezzoli, C. B. Mariani, A.L. DeStefano, C. Halter, J.F. Gusella, W.C. Nichols, R.H. Myers, T. Foroud, P.P.G. Investigators, Genomewide association study for onset age in Parkinson disease, *BMC Med. Genet.* 10 (2009).
- [19] T. Pillaiyar, S. Laufer, Kinases as potential therapeutic targets for anti-coronavirus therapy, *J. Med. Chem.* 65 (2022) 955–982.
- [20] B. Martinez-Gualda, D. Schols, S. De Jonghe, A patent review of adaptor associated kinase 1 (AAK1) inhibitors (2013–present), *Expert Opin. Ther. Pat.* 31 (2021) 911–936.
- [21] C. Gil, T. Ginex, I. Maestro, V. Nozal, L. Barrado-Gil, M.Á. Cuesta-Geijo, J. Urquiza, D. Ramirez, C. Alonso, N.E. Campillo, A. Martinez, COVID-19: drug targets and potential treatments, *J. Med. Chem.* 63 (2020) 12359–12386.
- [22] J. Stebbing, A. Phelan, I. Griffin, C. Tucker, O. Oechsle, D. Smith, P. Richardson, COVID-19: combining antiviral and anti-inflammatory treatments, *Lancet Infect. Dis.* 20 (2020) 400.
- [23] F. Cantini, L. Niccoli, D. Matarrese, E. Nicastri, P. Stobbione, D. Goletti, Baricitinib therapy in COVID-19: a pilot study on safety and clinical impact, *J. Infect.* 81 (2020) 318.

- [24] L. Cally, J.D. Druce, M.G. Catton, D.A. Jans, K.M. Wagstaff, The FDA-approved drug Ivermectin inhibits the replication of SARS-CoV-2 in vitro, *Antivir. Res.* 178 (2020), 104787.
- [25] S. Xia, M. Liu, C. Wang, W. Xu, Q. Lan, S. Feng, F. Qi, L. Bao, L. Du, S. Liu, C. Qin, F. Sun, Z. Shi, Y. Zhu, S. Jiang, L. Lu, Inhibition of SARS-CoV-2 (previously 2019-nCoV) infection by a highly potent pan-coronavirus fusion inhibitor targeting its spike protein that harbors a high capacity to mediate membrane fusion, *Cell Res.* 30 (2020) 343.
- [26] M. Puneekar, M. Kshirsagar, C. Tellapragada, K. Patil, Repurposing of antiviral drugs for COVID-19 and impact of repurposed drugs on the nervous system, *Microbial Pathogenesis* 168 (2022) 105608.
- [27] R. Raghuvanshi, S.B. Bharate, Recent developments in the use of kinase inhibitors for management of viral infections, *J. Med. Chem.* 65 (2022) 893–921.
- [28] E. Bekerman, G. Neveu, A. Shulla, J. Brannan, S.-Y. Pu, S. Wang, F. Xiao, R. Barouch-Bentov, R.R. Bakken, R. Mateo, J. Govero, C.M. Nagamine, M. S. Diamond, S. De Jonghe, P. Herdewijn, J.M. Dye, G. Randall, S. Einav, Anticancer kinase inhibitors impair intracellular viral trafficking and exert broad-spectrum antiviral effects, *J. Clin. Invest.* 127 (2017) 1338–1352.
- [29] W. Kostich, B.D. Hamman, Y.-W. Li, S. Naidu, K. Dandapani, J. Feng, A. Easton, C. Bourin, K. Baker, J. Allen, K. Savelieva, J.V. Louis, M. Dokania, S. Elavazhagan, P. Vattikundala, V. Sharma, M.L. Das, G. Shankar, A. Kumar, V.K. Holenarsipur, M. Gulianello, T. Molski, J.M. Brown, M. Lewis, Y. Huang, Y. Lu, R. Piesch, K. O'Malley, J. Lippy, A. Nouraldin, T.H. Lanthorn, G. Ye, A. Wilson, A. Balakrishnan, R. Denton, J.E. Grace, K.A. Lentz, K.S. Santone, Y. Bi, A. Main, J. Swaffield, K. Carson, S. Mandekar, R.K. Vikramadithyan, S.J. Nara, C. Dzierba, J. Bronson, J.E. Macor, R. Zaczek, R. Westphal, L. Kiss, L. Bristow, C.M. Conway, B. Zambrowicz, C.F. Albright, Inhibition of AAK1 kinase as a novel therapeutic approach to treat neuropathic pain, *J. Pharmacol. Exp. Therapeut.* 358 (2016) 371–386.
- [30] S. Verdonck, S.Y. Pu, F.J. Sorrell, J.M. Elkins, M. Froeyen, L.J. Gao, L.I. Prugar, D. E. Dorosky, J.M. Brannan, R. Barouch-Bentov, S. Knapp, J.M. Dye, P. Herdewijn, S. Einav, S. De Jonghe, Synthesis and structure–activity relationships of 3, 5-disubstituted-pyrrolo [2, 3-b] pyridines as inhibitors of adaptor-associated kinase 1 with antiviral activity, *J. Med. Chem.* 62 (2019) 5810.
- [31] C. Yuan, K. Ai, M. Xiang, C. Xie, M. Zhao, M. Wu, H. Li, Y. Wu, Y. Cao, C. Li, Y. Zhong, X. Pei, H.K.W. Law, L. Gao, Q. Xiao, X. Yang, Novel 1-hydroxy phenothiazinium-based derivative protects against bacterial sepsis by inhibiting AAK1-mediated LPS internalization and caspase-11 signaling, *Cell Death Dis.* 13 (2022) 722.
- [32] L. Farnung, S.M. Vos, C. Wigge, P. Cramer, Nucleosome–Chd1 structure and implications for chromatin remodelling, *Nature* 550 (2017) 539–542.
- [33] S. Rao, D. Gurbani, G. Du, R.A. Everley, C.M. Browne, A. Chaikwad, L. Tan, M. Schroeder, S. Gondi, S.B. Ficarro, T. Sim, N.D. Kim, M.J. Berberich, S. Knapp, J.A. Marto, K.D. Westover, P.K. Sorger, N.S. Gray, Leveraging compound promiscuity to identify targetable cysteines within the kinome, *Cell Chemical Biology* 26 (2019) 818(–+).
- [34] S.D. Conner, S.L. Schmid, The adaptor-associated kinase, AAK1, regulates cargo selection by AP2 complexes, *Mol. Biol. Cell* 13 (2002), 95A–95A.
- [35] J.J. Kim, R.E. Hibbs, Direct structural insights into GABAA receptor pharmacology, *Trends Biochem. Sci.* 46 (2021) 502–517.
- [36] D. Dorosky, L.I. Prugar, S. Pu, C. O'Brien, R. Bakken, S. De Jonghe, P. Herdewijn, J. Brannan, J.M. Dye, S. Einav, AAK1 and GAK inhibitors demonstrate activity against Filoviruses, *J. Immunol.* (2018) 200.
- [37] S. Pu, S. Schor, M. Karim, S. Saul, M. Robinson, S. Kumar, L.I. Prugar, D. E. Dorosky, J. Brannan, J.M. Dye, S. Einav, BIKE Regulates Dengue Virus Infection and Is a Cellular Target for Broad-Spectrum Antivirals, vol. 184, *Antiviral Research*, 2020.
- [38] K.R. Henry, K. D'Hondt, J.S. Chang, D.A. Nix, M. Cope, C.S.M. Chan, D.G. Drubin, S.K. Lemmon, The actin-regulating kinase Prk1p negatively regulates Scd5p, a suppressor of clathrin deficiency, in actin organization and endocytosis, *Curr. Biol.* 13 (2003) 1564–1569.
- [39] L. Kuai, S.-E. Ong, J.M. Madison, X. Wang, J.R. Duvall, T.A. Lewis, C.J. Luce, S. D. Conner, D.A. Pearlman, J.L. Wood, S.L. Schreiber, S.A. Carr, E.M. Scolnick, S. J. Haggarty, AAK1 identified as an inhibitor of neuregulin-1/ErbB4-dependent neurotrophic factor signaling using integrative chemical genomics and proteomics, *Chem. Biol.* 18 (2011) 891–906.
- [40] D. MacGrogan, J. Münch, J.L. de la Pompa, Notch and interacting signalling pathways in cardiac development, disease, and regeneration, *Nat. Rev. Cardiol.* 15 (2018) 685–704.
- [41] N. Gupta-Rossi, S. Ortica, V. Meas-Yedid, S. Heuss, J. Moretti, J.-C. Olivo-Marin, A. Israel, The adaptor-associated kinase 1, AAK1, is a positive regulator of the Notch pathway, *J. Biol. Chem.* 286 (2011) 18720–18730.
- [42] C. Chen, C.-H. Chu, Y. Chu, T.-Y. Chang, S.-W. Chen, S.-Y. Liang, Y.-C. Tsai, B.-C. Chen, H.-L. Tu, P.-L. Cheng, Neuronal paxillin and drebrin mediate BDNF-induced force transduction and growth cone turning in a soft-tissue-like environment, *Cell Rep.* 40 (2022), 111188.
- [43] M.J. Agajanian, M.P. Walker, A.D. Axtman, R.R. Ruela-de-Sousa, D.S. Serafin, A. D. Rabinowitz, D.M. Graham, M.B. Ryan, T. Tamir, Y. Nakamichi, M. V. Gammons, J.M. Bennett, R.M. Counago, D.H. Drewry, J.M. Elkins, C. Gileadi, O. Gileadi, P.H. Godoi, N. Kapadia, S. Mueller, A.S. Santiago, F.J. Sorrell, C. I. Wells, O. Fedorov, T.M. Willson, W.J. Zuercher, M.B. Major, WNT activates the AAK1 kinase to promote clathrin-mediated endocytosis of LRP6 and establish a negative feedback loop, *Cell Rep.* 26 (2019) 79 (+).
- [44] R. Nusse, H. Clevers, Wnt/ β -Catenin signaling, disease, and emerging therapeutic modalities, *Cell* 169 (2017) 985–999.
- [45] W.-K. Yang, Y.-H. Peng, H. Li, H.-C. Lin, Y.-C. Lin, T.-T. Lai, H. Suo, C.-H. Wang, W.-H. Lin, C.-Y. Ou, X. Zhou, H. Pi, Henry C. Chang, C.-T. Chien, Nak regulates localization of clathrin sites in higher-order dendrites to promote local dendrite growth, *Neuron* 72 (2011) 285–299.
- [46] I. Ferando, G.C. Faas, I. Mody, Diminished KCC2 confounds synapse specificity of LTP during senescence, *Nat. Neurosci.* 19 (2016) 1197–1200.
- [47] H.U. Zeilhofer, W.T. Ralvenius, M.A. Acuña, Chapter four - restoring the spinal pain gate: GABAA receptors as targets for novel analgesics, in: U. Rudolph (Ed.), *Advances in Pharmacology*, Academic Press, 2015, pp. 71–96.
- [48] W.A. Ray, C.P. Chung, K.T. Murray, K. Hall, C.M. Stein, Prescription of long-acting opioids and mortality in patients with chronic noncancer pain, *JAMA* 315 (2016) 2415–2423.
- [49] E.M. Hoffman, J.C. Watson, J. St Sauver, N.P. Staff, C.J. Klein, Association of long-term opioid therapy with functional status, adverse outcomes, and mortality among patients with polyneuropathy, *JAMA Neurol.* 74 (2017) 773–779.
- [50] C.J. Hayhurst, M.E. Durieux, Differential opioid tolerance and opioid-induced hyperalgesia: a clinical reality, *Anesthesiology* 124 (2016) 483–488.
- [51] D.B. Rosoff, G.D. Smith, F.W. Lohoff, Prescription opioid use and risk for major depressive disorder and anxiety and stress-related disorders: a multivariable mendelian randomization analysis, *JAMA Psychiatr.* 78 (2021) 151–160.
- [52] A.D. Kaye, E.M. Cornett, B. Hart, S. Patil, A. Pham, M. Spalitta, K.F. Mancuso, Novel pharmacological nonopioid therapies in chronic pain, *Curr. Pain Headache Rep.* (2018) 22.
- [53] Y. Fang, J. Wang, M. Zhao, Q. Zheng, C. Ren, Y. Wang, J. Zhang, Progress and challenges in targeted protein degradation for neurodegenerative disease therapy, *J. Med. Chem.* 65 (2022) 11454–11477.
- [54] S. Schmid, B. Jungwirth, V. Gehlert, M. Blobner, G. Schneider, S. Kratzer, K. Kellermann, G. Rammes, Intracerebroventricular injection of beta-amyloid in mice is associated with long-term cognitive impairment in the modified hole-board test, *Behav. Brain Res.* 324 (2017) 15–20.
- [55] A. Brice, α -Synuclein gene and Parkinson's disease, *Science* 279 (1998) 1113–1117.
- [56] K. Zhang, S. Zhu, J. Li, T. Jiang, L. Feng, J. Pei, G. Wang, L. Ouyang, B. Liu, Targeting autophagy using small-molecule compounds to improve potential therapy of Parkinson's disease, *Acta Pharm. Sin.* B 11 (2021) 3015.
- [57] Y. Zhang, M. James, F.A. Middleton, R.L. Davis, Transcriptional analysis of multiple brain regions in Parkinson's disease supports the involvement of specific protein processing, energy metabolism, and signaling pathways, and suggests novel disease mechanisms, *Am. J. Med. Genet. Part B: Neuropsychiatric Genetics* 137B (2005) 5–16.
- [58] D.R. Rosen, T. Siddique, D. Patterson, D.A. Figlewicz, P. Sapp, A. Hentati, D. Donaldson, J. Goto, J.P. O'Regan, H.-X. Deng, Z. Rahmani, A. Krizus, D. McKenna-Yasek, A. Cayabyab, S.M. Gaston, R. Berger, R.E. Tanzi, J. J. Halperin, B. Herzfeldt, R. Van den Bergh, W.-Y. Hung, T. Bird, G. Deng, D. W. Mulder, C. Smyth, N.G. Laing, E. Soriano, M.A. Pericak-Vance, J. Haines, G. A. Rouleau, J.S. Gusella, H.R. Horvitz, R.H. Brown, Mutations in Cu/Zn superoxide dismutase gene are associated with familial amyotrophic lateral sclerosis, *Nature* 362 (1993) 59–62.
- [59] A. Baldwin, W. Li, M. Grace, J. Pearlberg, E. Harlow, K. Münger, D. A. Grueneberg, Kinase requirements in human cells: II. Genetic interaction screens identify kinase requirements following HPV16 E7 expression in cancer cells, *Proc. Natl. Acad. Sci. USA* 105 (2008) 16478–16483.
- [60] J.M. White, G.R. Whittaker, Fusion of enveloped viruses in endosomes, *Traffic* 17 (2016) 593–614.
- [61] D. Eshel, I. Miyara, T. Ailing, A. Dinoor, D. Prusky, pH regulates endoglucanase expression and virulence of *Alternaria alternata* persimmon fruit, *Mol. Plant Microbe Interact.* 15 (2002) 774–779.
- [62] P. Richardson, I. Griffin, C. Tucker, D. Smith, O. Oechsle, A. Phelan, M. Rawling, E. Savory, J. Stebbing, Baricitinib as potential treatment for 2019-nCoV acute respiratory disease, *Lancet* 395 (2020) e30–e31.
- [63] R. Lu, X. Zhao, J. Li, P. Niu, B. Yang, H. Wu, W. Wang, H. Song, B. Huang, N. Zhu, Y. Bi, X. Ma, F. Zhan, L. Wang, T. Hu, H. Zhou, Z. Hu, W. Zhou, L. Zhao, J. Chen, Y. Meng, J. Wang, Y. Lin, J. Yuan, Z. Xie, J. Ma, W.J. Liu, D. Wang, W. Xu, E. C. Holmes, G.F. Gao, G. Wu, W. Chen, W. Shi, W. Tan, Genomic characterisation and epidemiology of 2019 novel coronavirus: implications for virus origins and receptor binding, *Lancet* 395 (2020) 565–574.
- [64] B. Tang, J. Zhu, Y. Cong, W. Yang, C. Kong, W. Chen, Y. Wang, Y. Zeng, J. Ji, The landscape of coronavirus disease 2019 (COVID-19) and integrated analysis SARS-CoV-2 receptors and potential inhibitors in lung adenocarcinoma patients, *Front. Cell Dev. Biol.* 8 (2020).
- [65] G. Moreno-Gonzalez, A. Mussetti, A. Albasanz-Puig, I. Salvador, A. Sureda, C. Gudiol, R. Salazar, M. Marin, M. Garcia, V. Navarro, I. de la Haba Vaca, E. Coma, G. Sanz-Linares, X. Dura, S. Fontanals, G. Serrano, C. Cruz, R. Manez, A Phase I/II Clinical Trial to evaluate the efficacy of baricitinib to prevent respiratory insufficiency progression in onco-hematological patients affected with COVID19: a structured summary of a study protocol for a randomised controlled trial, *Trials* (2021) 22.
- [66] J. Huang, C. Zhou, J. Deng, J. Zhou, JAK inhibition as a new treatment strategy for patients with COVID-19, *Biochem. Pharmacol.* 202 (2022), 115162–115162.
- [67] G. Neveu, A. Ziv-Av, R. Barouch-Bentov, E. Berkerman, J. Mulholland, S. Einav, AAK1 and GAK regulate hepatitis C virus entry and are potential drug targets, *J. Virol.* 89 (2015) 4387.
- [68] G. Neveu, A. Ziv-Av, R. Barouch-Bentov, E. Berkerman, J. Mulholland, S. Einav, AP-2-Associated protein kinase 1 and cyclin G-associated kinase regulate hepatitis C virus entry and are potential drug targets, *J. Virol.* 89 (2015) 4387–4404.

- [69] J. Diao, H. Pantua, H. Ngu, L. Komuves, L. Diehl, G. Schaefer, S.B. Kapadia, Hepatitis C virus induces epidermal growth factor receptor activation via CD81 binding for viral internalization and entry, *J. Virol.* 86 (2012) 10935–10949.
- [70] C. Wang, J. Wang, L. Shuai, X. Ma, H. Zhang, R. Liu, W. Chen, X. Wang, J. Ge, Z. Wen, Z. Bu, The serine/threonine kinase AP2-associated kinase 1 plays an important role in rabies virus entry, *Viruses-Basel* 12 (2020).
- [71] J. Luo, Y. Zhang, Y. Wang, Q. Liu, L. Chen, B. Zhang, Y. Luo, S. Huang, X. Guo, Rhabdovirus infection is dependent on serine/threonine kinase AP2-associated kinase 1, *Life* 10 (2020) 170.
- [72] S.Y. Pu, F. Xiao, S. Schor, E. Bekerman, F. Zanini, R. Barouch-Bentov, C. M. Nagamine, S. Einav, Feasibility and biological rationale of repurposing sunitinib and erlotinib for dengue treatment, *Antivir. Res.* 155 (2018) 67.
- [73] E. Bekerman, G. Neveu, A. Shulla, J. Brannan, S.Y. Pu, S. Wang, F. Xiao, R. Barouch-Bentov, R.R. Bakken, R. Mateo, J. Govero, C.M. Nagamine, M. S. Diamond, S. De Jonghe, P. Herdewijn, J.M. Dye, G. Randall, S. Einav, Anticancer kinase inhibitors impair intracellular viral trafficking and exert broad-spectrum antiviral effects, *J. Clin. Invest.* 127 (2017) 1338.
- [74] L.M. Rogers, Z. Wang, S.L. Mott, A.J. Dupuy, G.J. Weiner, A genetic screen to identify gain- and loss-of-function modifications that enhance T-cell infiltration into tumors, *Cancer Immunol. Res.* 8 (2020) 1206–1214.
- [75] T. Pan, Y. He, H. Chen, J. Pei, Y. Li, R. Zeng, J. Xia, Y. Zuo, L. Qin, S. Chen, L. Xiao, H. Zhou, Identification and validation of a prognostic gene signature for diffuse large B-cell lymphoma based on tumor microenvironment-related genes, *Front. Oncol.* 11 (2021).
- [76] T. Liu, Y. Wang, J. Wang, C. Ren, H. Chen, J. Zhang, DYRK1A inhibitors for disease therapy: current status and perspectives, *Eur. J. Med. Chem.* 229 (2022), 114062.
- [77] Z. Zhao, H. Wu, L. Wang, Y. Liu, S. Knapp, Q. Liu, N.S. Gray, Exploration of type II binding mode: a privileged approach for kinase inhibitor focused drug discovery? *ACS Chem. Biol.* 9 (2014) 1230–1241.
- [78] Y. Liu, N.S. Gray, Rational design of inhibitors that bind to inactive kinase conformations, *Nat. Chem. Biol.* 2 (2006) 358–364.
- [79] J. Zhang, L. Zhang, J. Wang, L. Ouyang, Y. Wang, Polo-like kinase 1 inhibitors in human cancer therapy: development and therapeutic potential, *J. Med. Chem.* 65 (2022) 10133–10160.
- [80] R.A. Hartz, V.T. Ahuja, S.J. Nara, C.M.V. Kumar, J.M. Brown, L.J. Bristow, R. Rajamani, J.K. Muckelbauer, D. Camac, S.E. Kiefer, L. Hunihan, M. Gulianello, M. Lewis, A. Easton, J.S. Lippy, N. Surti, S.N. Pattipati, M. Dokania, S. Elavazhagan, K. Dandapani, B.D. Hamman, J. Allen, W. Kostich, J.J. Bronson, J.E. Macor, C.D. Dzierba, Discovery, structure-activity relationships, and in vivo evaluation of novel aryl amides as brain penetrant adaptor protein 2-associated kinase 1 (AAK1) inhibitors for the treatment of neuropathic pain, *J. Med. Chem.* 64 (2021) 11090–11128.
- [81] G. Luo, L. Chen, W.A. Kostich, B. Hamman, J. Allen, A. Easton, C. Bourin, M. Gulianello, J. Lippy, S. Nara, T.K. Maishal, K. Thiyagarajan, P. Jalagam, S. N. Pattipati, K. Dandapani, M. Dokania, P. Vattikundala, V. Sharma, S. Elavazhagan, M.K. Verma, M.L. Das, S. Wagh, A. Balakrishnan, B.M. Johnson, K.S. Santone, G. Thalody, R. Denton, H. Saminathan, V.K. Holenarsipur, A. Kumar, A. Rao, S.P. Putlur, S.K. Sarvasiddhi, G. Shankar, J.V. Louis, M. Ramarao, C.M. Conway, Y.-W. Li, R. Pieschl, Y. Tian, Y. Hong, J. Ditta, A. Mathur, J. Li, D. Smith, J. Pawluczyk, D. Sun, S. Yip, D.-R. Wu, M. Vetrivelvan, A. Gupta, A. Wilson, S. Gopinathan, S. Wason, L. Bristow, C. F. Albright, J.J. Bronson, J.E. Macor, C.D. Dzierba, Discovery of (S)-1-((2',6-Bis (difluoromethyl)-2,4'-bipyridin-5-yl)oxy)-2,4-dimethylpentan-2-amine (BMS-986176/LX-9211): A Highly selective, CNS penetrable, and orally active AdaptorProtein-2 associated kinase 1 inhibitor in clinical trials for the treatment of neuropathic pain, *J. Med. Chem.* 65 (2022) 4457–4480.
- [82] M.W. Karaman, S. Herrgard, D.K. Treiber, P. Gallant, C.E. Atteridge, B. T. Campbell, K.W. Chan, P. Ciceri, M.I. Davis, P.T. Edeen, R. Faraoni, M. Floyd, J. P. Hunt, D.J. Lockhart, Z.V. Milanov, M.J. Morrison, G. Pallares, H.K. Patel, S. Pritchard, L.M. Wodicka, P.P. Zarrinkar, A quantitative analysis of kinase inhibitor selectivity, *Nat. Biotechnol.* 26 (2008) 127–132.
- [83] G.J. Bennett, Y.-K. Xie, A peripheral mononeuropathy in rat that produces disorders of pain sensation like those seen in man, *Pain* 33 (1988) 87–107.
- [84] T.T. Wager, R.Y. Chandrasekaran, X. Hou, M.D. Troutman, P.R. Verhoest, A. Villalobos, Y. Will, Defining desirable central nervous system drug space through the alignment of molecular properties, in vitro ADME, and safety attributes, *ACS Chem. Neurosci.* 1 (2010) 420–434.
- [85] G. Luo, L. Chen, W.A. Kostich, B. Hamman, J. Allen, A. Easton, C. Bourin, M. Gulianello, J. Lippy, S. Nara, S.N. Pattipati, K. Dandapani, M. Dokania, P. Vattikundala, V. Sharma, S. Elavazhagan, M.K. Verma, M.L. Das, S. Wagh, A. Balakrishnan, B.M. Johnson, K.S. Santone, G. Thalody, R. Denton, H. Saminathan, V.K. Holenarsipur, A. Kumar, A. Rao, S.P. Putlur, S. K. Sarvasiddhi, G. Shankar, J.V. Louis, M. Ramarao, C.M. Conway, Y.-W. Li, R. Pieschl, Y. Tian, Y. Hong, L. Bristow, C.F. Albright, J.J. Bronson, J.E. Macor, C. D. Dzierba, Discovery and optimization of biaryl alkyl ethers as a novel class of highly selective, CNS-penetrable, and orally active AdaptorProtein-2-associated kinase 1 (AAK1) inhibitors for the Potential Treatment of neuropathic pain, *J. Med. Chem.* 65 (2022) 4534–4564.
- [86] P. Eastwood, J. Gonzalez, S. Paredes, S. Fonquerna, A. Cardús, J.A. Alonso, A. Nueda, T. Domenech, R.F. Reinoso, B. Vidal, Discovery of potent and selective bicyclic A2B adenosine receptor antagonists via bioisosteric amide replacement, *Bioorg. Med. Chem. Lett* 20 (2010) 1634–1637.
- [87] S. Sun, Q. Jia, Z. Zhang, Applications of amide isosteres in medicinal chemistry, *Bioorg. Med. Chem. Lett* 29 (2019) 2535–2550.
- [88] R.A. Hartz, V.T. Ahuja, S.J. Nara, C.M.V. Kumar, R.K.V.L.P. Manepalli, S. K. Sarvasiddhi, S. Honkhambe, V. Patankar, B. Dasgupta, R. Rajamani, J. K. Muckelbauer, D.M. Camac, G. Ghosh, M. Pokross, S.E. Kiefer, J.M. Brown, L. Hunihan, M. Gulianello, M. Lewis, J.S. Lippy, N. Surti, B.D. Hamman, J. Allen, W.A. Kostich, J.J. Bronson, J.E. Macor, C.D. Dzierba, Bicyclic heterocyclic replacement of an aryl amide leading to potent and kinase-selective adaptor protein 2-associated kinase 1 inhibitors, *J. Med. Chem.* 65 (2022) 4121–4155.
- [89] J.V. Louis, Y. Lu, R. Pieschl, Y. Tian, Y. Hong, K. Dandapani, S. Naidu, R. K. Vikramadithyan, C. Dzierba, S.K. Sarvasiddhi, S.J. Nara, J. Bronson, J. E. Macor, C. Albright, W. Kostich, Y.-W. Li, H-3 BMT-046091 a potent and selective radioligand to determine AAK1 distribution and target engagement, *Neuropharmacology* 118 (2017) 167–174.
- [90] E. Bekerman, S. Einav, Combating emerging viral threats, *Science* 348 (2015) 282–283.
- [91] P. Bamborough, D. Drewry, G. Harper, G.K. Smith, K. Schneider, Assessment of chemical coverage of kinome space and its implications for kinase drug discovery, *J. Med. Chem.* 51 (2008) 7898–7914.
- [92] S. Verdonck, S.-Y. Pu, F.J. Sorrell, J.M. Elkins, M. Froeyen, L.-J. Gao, L.I. Prugar, D.E. Dorosky, J.M. Brannan, R. Barouch-Bentov, S. Knapp, J.M. Dye, P. Herdewijn, S. Einav, S. De Jonghe, Synthesis and structure-activity relationships of 3,5-Disubstituted-pyrrolo-2,3-b pyridines as inhibitors of adaptor-associated kinase 1 with antiviral activity, *J. Med. Chem.* 62 (2019) 5810–5831.
- [93] D. Cunningham, E. Plow, D. Allexandre, B.I. Rini, M.P. Davis, Neurophysiological changes: sunitinib fatigue severity, *J. Clin. Oncol.* 34 (2016), e21683-e21683.
- [94] M.C. Genovese, J. Kremer, O. Zamani, C. Ludovico, M. Krogulec, L. Xie, S. D. Beattie, A.E. Koch, T.E. Cardillo, T.P. Rooney, W.L. Macias, S. de Bono, D. E. Schlichting, J.S. Smolen, Baricitinib in patients with refractory rheumatoid arthritis, *N. Engl. J. Med.* 374 (2016) 1243–1252.
- [95] T.N. Hoang, M. Pino, A.K. Boddapati, E.G. Viox, C.E. Starke, A.A. Upadhyay, S. Gumber, K. Busman-Sahay, Z. Strongin, J.L. Harper, G.K. Tharp, K.L. Pellegrini, S. Kirejczyk, K. Zandi, S. Tao, T.R. Horton, E.N. Beagle, E.A. Mahar, M.Y. Lee, J. Cohen, S.M. Jean, J.S. Wood, F. Connor-Stroud, R.L. Stammen, O.M. Delmas, S. Wang, K.A. Cooney, M.N. Sayegh, L. Wang, D. Weiskopf, P.D. Filev, J. Waggoner, A. Piantadosi, S.P. Kasturi, H.A. Shakhshir, S.P. Ribeiro, R.P. Sekaly, R.D. Levit, J.D. Estes, T.H. Vanderford, R.F. Schinazi, S.E. Bosinger, M. Paiardini, Baricitinib treatment resolves lower airway inflammation and neutrophil recruitment in SARS-CoV-2-infected rhesus macaques, *Cell* 184 (2021) 460.
- [96] Fedratinib becomes new option in myelofibrosis, *Cancer Discov.* 9 (2019), 1332–1332.
- [97] P. Kar, M.M. Saleh-E-In, N. Jaishee, A. Anandraj, E. Kormuth, B. Vellingiri, C. Angione, P.K.S.M. Rahman, S. Pillay, A. Sen, D. Naidoo, A. Roy, Y.E. Choi, Computational profiling of natural compounds as promising inhibitors against the spike proteins of SARS-CoV-2 wild-type and the variants of concern, viral cell-entry process, and cytokine storm in COVID-19, *J. Cell. Biochem.* 123 (2022) 964–986.
- [98] F. Bu, J. Zhang, W. Shuai, J. Liu, Q. Sun, L. Ouyang, Repurposing drugs in autophagy for the treatment of cancer: from bench to bedside, *Drug Discov. Today* 27 (2022) 1815–1831.
- [99] J. Baselga, Targeting tyrosine kinases in cancer: the second wave, *Science* 312 (2006) 1175–1178.
- [100] J.S. Sebolt-Leopold, J.M. English, Mechanisms of drug inhibition of signalling molecules, *Nature* 441 (2006) 457–462.
- [101] K.-H. Lee, S.-E. Kim, Y.-S. Lee, SP600125, a selective JNK inhibitor, aggravates hepatic ischemia-reperfusion injury, *Exp. Mol. Med.* 38 (2006) 408–416.
- [102] S. Galimberti, M. Petrini, C. Barate, F. Ricci, S. Balducci, S. Grassi, F. Guerrini, E. Ciabatti, S. Mechelli, A. Di Paolo, C. Baldini, L. Baglietto, L. Macera, P. G. Spezia, F. Maggi, Tyrosine kinase inhibitors play an antiviral action in patients affected by chronic myeloid leukemia: a possible model supporting their use in the fight against SARS-CoV-2, *Front. Oncol.* 10 (2020) 1428.
- [103] Sivapriya K. Vanaja, Ashley J. Russo, B. Behl, I. Banerjee, M. Yankova, Sachin D. Deshmukh, Vijay A.K. Rathinam, Bacterial outer membrane vesicles mediate cytosolic localization of LPS and caspase-11 activation, *Cell* 165 (2016) 1106–1119.
- [104] Y. Achoui, I.A. Leaf, J.A. Hagar, M.F. Fontana, C.G. Campos, D.E. Zak, M.H. Tan, P.A. Cotter, R.E. Vance, A. Aderem, E.A. Miao, Caspase-11 protects against bacteria that escape the vacuole, *Science* 339 (2013) 975–978.
- [105] G.A. Ellestad, (5Z)-7-Oxozeaenol: a novel and potent resorcylic acid lactone kinase inhibitor with a cis-enone Michael acceptor, *Chirality* 31 (2019) 110–117.
- [106] J. Rani, S. Kumar, M. Saini, J. Mundlia, P.K. Verma, Biological potential of pyrimidine derivatives in a new era, *Res. Chem. Intermed.* 42 (2016) 6777–6804.
- [107] D.H. Drewry, J.K. Annor-Gyamfi, C.I. Wells, J.E. Pickett, V. Dederer, F. Preuss, S. Mathea, A.D. Axtman, Identification of pyrimidine-based lead compounds for understudied kinases implicated in driving neurodegeneration, *J. Med. Chem.* 65 (2022) 1313–1328.
- [108] M. Dammann, M. Kramer, M.O. Zimmermann, F.M. Boeckler, Quadruple target evaluation of diversity-optimized halogen-enriched fragments (HEFLibs) reveals substantial ligand efficiency for AP2-associated protein kinase 1 (AAK1), *Front. Chem.* 9 (2022).
- [109] J.M. Elkins, V. Fedele, M. Szklarz, K.R.A. Azeez, E. Salah, J. Mikolajczyk, S. Romanov, N. Sepetov, X.-P. Huang, B.L. Roth, A.A.H. Zen, D. Fourches, E. Muratov, A. Tropsha, J. Morris, B.A. Teicher, M. Kunkel, E. Polley, K.E. Lackey, F.L. Atkinson, J.P. Overington, P. Bamborough, S. Mueller, D.J. Price, T. M. Willson, D.H. Drewry, S. Knapp, W.J. Zuercher, Comprehensive characterization of the published kinase inhibitor set, *Nat. Biotechnol.* 34 (2016) 95–103.

- [110] C. Wells, R.M. Counago, J.C. Limas, T.L. Almeida, J.G. Cook, D.H. Drewry, J. M. Elkins, O. Gileadi, N.R. Kapadia, A. Lorente-Macias, J.E. Pickett, A. Riemen, R. R. Ruela-de-Sousa, T.M. Willson, C. Zhang, W.J. Zuercher, R. Zutshi, A. D. Axtman, SGC-AAK1-1: a chemical probe targeting AAK1 and BMP2K, *ACS Med. Chem. Lett.* 11 (2020) 340–345.
- [111] R.A.M. Serafim, J.M. Elkins, W.J. Zuercher, S.A. Laufer, M. Gehringer, Chemical probes for understudied kinases: challenges and opportunities, *J. Med. Chem.* 65 (2022) 1132–1170.
- [112] V.M. Shahani, D.P. Ball, A.V. Ramos, Z. Li, P.A. Spagnuolo, S. Haftchenary, A. D. Schimmer, S. Trudel, P.T. Gunning, A 2,6,9-hetero-trisubstituted purine inhibitor exhibits potent biological effects against multiple myeloma cells, *Bioorg. Med. Chem.* 21 (2013) 5618–5628.
- [113] R. He, B. Xu, L. Ping, X. Lv, Structural optimization towards promising β -methyl-4-acrylamido quinoline derivatives as PI3K/mTOR dual inhibitors for anti-cancer therapy: the in vitro and in vivo biological evaluation, *Eur. J. Med. Chem.* 214 (2021), 113249.
- [114] H.C. Mefford, Expanding role of GABAA receptors in generalised epilepsies, *Lancet Neurol.* 17 (2018) 657–658.
- [115] S. Masiulis, R. Desai, T. Uchański, I. Serna Martin, D. Laverty, D. Karia, T. Malinauskas, J. Zivanov, E. Pardon, A. Kotecha, J. Steyaert, K.W. Miller, A. R. Aricescu, GABAA receptor signalling mechanisms revealed by structural pharmacology, *Nature* 565 (2019) 454–459.
- [116] M. Keller, F. Dubois, S. Teulier, A.P.J. Martin, J. Levallet, E. Maille, S. Brosseau, N. Elie, A. Hergovich, E. Bergot, J. Camonis, G. Zalcman, G. Levallet, NDR2 kinase contributes to cell invasion and cytokinesis defects induced by the inactivation of RASSF1A tumor-suppressor gene in lung cancer cells, *J. Exp. Clin. Cancer Res.* 38 (2019) 158.
- [117] A.A.D. Sharif, A. Hergovich, The NDR/LATS protein kinases in immunology and cancer biology, *Semin. Cancer Biol.* 48 (2018) 104–114.
- [118] B.A. Feinberg, J. Wojtynek, I. Dokubo, Y. Jeune-Smith, J. Kish, A. Gajra, Prescribing preferences for hormone sensitive (HR+) metastatic breast cancer (mBC) in the CDK 4/6 inhibitor (CDK 4/6i) era, *J. Clin. Oncol.* 38 (2020), e13058–e13058.
- [119] M. Bury, B. Le Calvé, G. Ferbeyre, V. Blank, F. Lessard, New insights into CDK regulators: novel opportunities for cancer therapy, *Trends Cell Biol.* 31 (2021) 331–344.
- [120] B. Zhou, W. Lin, Y. Long, Y. Yang, H. Zhang, K. Wu, Q. Chu, Notch signaling pathway: architecture, disease, and therapeutics, *Signal Transduct. Targeted Ther.* 7 (2022) 95.
- [121] E.Y. Rim, H. Clevers, R. Nusse, The wnt pathway: from signaling mechanisms to synthetic modulators, *Annu. Rev. Biochem.* 91 (2022) 571–598.
- [122] S.G. Pai, B.A. Carneiro, J.M. Mota, R. Costa, C.A. Leite, R. Barroso-Sousa, J. B. Kaplan, Y.K. Chae, F.J. Giles, Wnt/beta-catenin pathway: modulating anticancer immune response, *J. Hematol. Oncol.* 10 (2017) 101.
- [123] H. Xiang, J. Zhang, C. Lin, L. Zhang, B. Liu, L. Ouyang, Targeting autophagy-related protein kinases for potential therapeutic purpose, *Acta Pharm. Sin. B* 10 (2020) 569–581.
- [124] C. Sarah, Targeting host kinases, *Nat. Rev. Drug Discov.* 16 (2017), 314–314.
- [125] F. Cantini, L. Niccoli, D. Matarrese, E. Nicastri, P. Stobbione, D. Goletti, Baricitinib therapy in COVID-19: a pilot study on safety and clinical impact, *J. Infect.* 81 (2020) 318–356.
- [126] L. Colloca, T. Ludman, D. Bouhassira, R. Baron, A.H. Dickenson, D. Yarnitsky, R. Freeman, A. Truini, N. Attal, N.B. Finnerup, C. Eccleston, E. Kalso, D. L. Bennett, R.H. Dworkin, S.N. Raja, Neuropathic pain, *Nat. Rev. Dis. Prim.* 3 (2017), 17002.
- [127] K. Bannister, J. Sachau, R. Baron, A.H. Dickenson, Neuropathic pain: mechanism-based therapeutics, *Annu. Rev. Pharmacol. Toxicol.* 60 (2020) 257–274.
- [128] N.D. Marino, J.Y. Zhang, A.L. Borges, A.A. Sousa, L.M. Leon, B.J. Rauch, R. T. Walton, J.D. Berry, J.K. Joung, B.P. Kleinstiver, J. Bondy-Denomy, Discovery of widespread type I and type V CRISPR-Cas inhibitors, *Science* 362 (2018) 240–242.
- [129] L.M. Jones, K. Melgar, L. Bolanos, K. Hueneman, M.M. Walker, J.-K. Jiang, K. M. Wilson, X. Zhang, J. Shen, F. Jiang, P. Sutter, A. Wang, X. Xu, G.J. Tawa, S. B. Hoyt, M. Wunderlich, E. O'Brien, J.P. Perentesis, D.T. Starczynowski, C. J. Thomas, Targeting AML-associated FLT3 mutations with a type I kinase inhibitor, *J. Clin. Investig.* 130 (2020) 2017–2023.
- [130] R. Morishita, J. Higaki, S.I. Hayashi, Y. Yo, M. Aoki, S. Nakamura, A. Moriguchi, H. Matsushita, K. Matsumoto, T. Nakamura, T. Ogihara, Role of hepatocyte growth factor in endothelial regulation: prevention of high D-glucose-induced endothelial cell death by prostaglandins and phosphodiesterase type 3 inhibitor, *Diabetologia* 40 (1997) 1053–1061.
- [131] G.L. Goebel, X. Qiu, P. Wu, Kinase-targeting small-molecule inhibitors and emerging bifunctional molecules, *Trends Pharmacol. Sci.* 43 (2022) 866–881.
- [132] K.K. Chan, A.R. Bass, Checkpoint inhibitor-induced polymyalgia rheumatica controlled by cobimetinib, a MEK 1/2 inhibitor, *Ann. Rheum. Dis.* 78 (2019) e70.
- [133] B. Wang, H. Wu, C. Hu, H. Wang, J. Liu, W. Wang, Q. Liu, An overview of kinase downregulators and recent advances in discovery approaches, *Signal Transduct. Targeted Ther.* 6 (2021) 423.
- [134] X. Lu, J.B. Smaill, K. Ding, New promise and opportunities for allosteric kinase inhibitors, *Angew. Chem. Int. Ed.* 59 (2020) 13764–13776.
- [135] A. Abdeldayem, Y.S. Raouf, S.N. Constantinescu, R. Moriggi, P.T. Gunning, Advances in covalent kinase inhibitors, *Chem. Soc. Rev.* 49 (2020) 2617–2687.
- [136] W.F. Kean, H.E. Howard-Lock, C.J.L. Lock, Chirality in antirheumatic drugs, *Lancet* 338 (1991) 1565–1568.
- [137] T. Rodrigues, D. Reker, P. Schneider, G. Schneider, Counting on natural products for drug design, *Nat. Chem.* 8 (2016) 531–541.